Mathmatical Analysis Zorich Comprehensions

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Contents

Figure 1: One week, one Zorich

1 Chapter2. The Real Numbers

1.1 Comprehensions

The main content covers the basic properties of real numbers, the definitions and properties of rational and irrational numbers, natural numbers and mathematical induction, and, most importantly, several equivalence principles describing the completeness of real numbers (more generally, the density of real numbers). (Zorich calls these fundamental lemmas, which serve as the foundation for constructing real number theory, but calling them principles is fine.) Finally, some information on Cantor's set theory is added.

The method of describing the completeness of real numbers using Didekind partitioning, which we learned in Fehkingoltz's "Calculus Tutorial (I)" can also be incorporated into this.

Basic Lemmas of the Real Number Completeness Theorem

1. (The least upper (lower) bound principle) Every non-empty set of real numbers that is bounded from above (below) has a unique upper (lower) bound.
2. (Cauchy-Cantor Principle) (The nested interval lemma) The set of closed intervals on the real number axis whose length $\to 0$ determines a unique point (a common point that belongs to all these closed intervals).
3. (Borel-Lebesgue Principle) (The finite covering lemma) Every system of open intervals covering a closed interval contains a finite subcovering.
4. (The Bolzano-Weierstrass Principle) (The limit point lemma) Every bounded infinite set of real numbers has at least one limit point.

Notes. 1. If the length of the closed intervals is not approaching 0, there are still existing common points, but they are not unique.
2. If infinite is limit point, then the Bolzano-Weierstrass Principle is suit for unbounded infinite set of real numbers.

2 important properties of natural numbers $\mathbb{Q}$.

2 Important Properties of Natural Numbers

Thm1. (The Foundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$, it could be represented uniquely as: $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where p_j are different prime numbers, α_j are positive integers.

Thm2. (The Principle of Archimedes) Given $h > 0$, then $\forall x \in \mathbb{R}$, $\exists$ unique $k \in \mathbb{Z}$, s.t. $(k-1)h \leq x \leq kh$.

Cantor Set Theory.

Countable and Uncountable Sets

Proposition 1. card(Countable Set)=card($\mathbb{N}$)=$\aleph_0$, card(Uncountable Set)=card($\mathbb{R}$)=$\aleph_1$ (cardinality of the continuum).

Proposition 2. Any subset of a at most countable set (card(X)$\leq$card($\mathbb{N}$)) is at most countable; the at most countable union of at most countable sets is at most countable.

Proposition 3. (Corollary) The direct sum (or direct product) of countable sets is countable. (Corollary) The set of natural numbers or algebraic numbers is countable.

Thm 1. (Cantor's Theorem) card($\mathbb{N}$) $<$ card($\mathbb{R}$).

Proposition 4. (Corollary) the irrational numbers or transcendental numbers exisst, and the set of all of them is uncountable.

1.2 Problems Sets

An interesting example-the Cantor set

Example. Let $x \in [0,1] \subset \mathbb{R}$, its ternary expansion is $x = 0.a_1a_2a_3\cdots$ ($a_i \in \{0,1,2\}$) satisfying: $\forall i \in \mathbb{N}, a \neq 1$, then the set of all such elements is called the Cantor set.

Notes. The Cantor set is uncountable (It's easy: the elements of it can be onto mapped to all binary decimals on $[0,1]$), and we'll then learn: Cantor set is measure zero in sense of Lebesgue.

2 2 Classical Limits

2.1 Original

2.1.1 $\lim_{n\to\infty} \frac{a^n}{n^k}$

The problem is:

$$\lim_{n\to\infty} \frac{a^n}{n^k} \quad \text{where } a > 1,\ k > 0$$

Solution (Solution1). Based on **Binomial Theorem**
$a^n = [a \triangleq 1 + \lambda] = 1 + n\lambda + \frac{n(n-1)}{2}\lambda^2 + \cdots > \frac{n(n-1)}{2}\lambda^2$
also $n - 1 > \frac{n}{2},\ as\ n > 2$, then $a^n > \frac{n^2}{4}(a-1)^2$.
As $k = 1$, $\lim_{n\to\infty} \frac{a^n}{n} = \infty$, and as $k > 1$, $\lim_{n\to\infty} \frac{a^n}{n^k} = \lim_{n\to\infty}(\frac{(a^{\frac{1}{k}})^n}{n})^k = \infty$.

Solution (Solution2). Based on **L'Hospital's Rule**
$\lim_{n\to\infty} \frac{a^n}{n^k} = \lim_{n\to\infty} \frac{\ln a \cdot a^n}{k \cdot n^{k-1}} = \lim_{n\to\infty} \frac{(\ln a)^2 \cdot a^n}{k(k-1)\cdot n^{k-2}} = \cdots = \lim_{n\to\infty} \frac{(\ln a)^k \cdot a^n}{k!} = \infty$.

Solution (Solution3). (Haven't found a totally different solution yet.)

2.1.2 $\lim_{n\to\infty} n^{\frac{1}{n}}$

The problem is:

$$\lim_{n\to\infty} n^{\frac{1}{n}}$$

Solution (Solution1). Due to 1. $a^n > \frac{n^2}{4}(a-1)^2$, let $a = n^{\frac{1}{n}}$, then $0 < n^{\frac{1}{n}} < \frac{2}{\sqrt{n}}$. Thus $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

Solution (Solution2). Based on Mean Value Inequalities Chain(GM-AM): $n^{\frac{1}{n}} = (\sqrt{n} \cdot \sqrt{n} \cdot 1 \cdots 1)^{\frac{1}{n}} \le \frac{2\sqrt{n}+n-2}{n}$, then $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

2.2 Some repurposed variants

2.2.1 $\lim_{n\to+\infty} \sqrt[n]{n!}$

The problem is:

$$\lim_{n\to+\infty} \sqrt[n]{n!}$$

Solution (Solution1). Based on **Stirling's Approximation**: $n! \sim \sqrt{2\pi n}(\frac{n}{e})^n$, then $\sqrt[n]{n!} \sim \sqrt[n]{\sqrt{2\pi n}}(\frac{n}{e}) = \frac{n}{e} \cdot (2\pi n)^{\frac{1}{2n}}$. Thus $\lim_{n\to+\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution2). $n! \geq \left(\frac{n}{2}\right)^{\frac{n}{2}}$ because half of the factors are at least $\frac{n}{2}$

Solution (Solution3). Consider Taylor series, $e^x \geq \frac{x^n}{n!}$, $\forall x \geq 0, n \in \mathbb{N}$, then $n! \geq \frac{x^n}{e^x}$, take $x = n$ problem solved.

Solution (Solution4). Based on **Stolze Cesaro Theorem**: $\lim\limits_{n\to+\infty} \sqrt[n]{n!} = \lim\limits_{n\to+\infty} \frac{n!^{\frac{1}{n}}}{1} = \lim\limits_{n\to+\infty} \frac{\ln(n!)}{n} = \lim_{n\to+\infty} \frac{\ln(n)+\ln(n-1)+\cdots+\ln(1)}{n}$, then apply Stolze Cessaro Theorem, $\lim\limits_{n\to+\infty} \frac{\ln(n)+\ln(n-1)+\cdots+\ln(1)}{n} = \lim\limits_{n\to+\infty} \frac{\ln(n+1)}{1} =$, then $\lim\limits_{n\to+\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution5). Using a multiplicative variant of Gauss' trick we get: $(n!)^2 = (1 \cdot n)(2 \cdot (n-1)) \cdots ((n-1) \cdot 2)(n \cdot 1) \geq n^n$.

3 Chapter 3.Limits

3.1 Comprehensions

3.1.1 ξ.1 The Limit of a Sequence

First, the most basic is the ε-N definition of sequence limits and the basic properties of limits (existence $\implies$ boundedness). On this basis, the operational processes of limits are established (four arithmetic operations and inequality relations, note: the strictness of inequalities may not be maintained after the limit process). Then, we proceed to the discussion of the existence of limits (introducing two types of convergence criteria: the /general/ Cauchy convergence criterion and the /special - monotonic sequence/ Weierstrass theorem), and an important method: subsequences, subsequence limits, and upper and lower limits.

Furthermore, this section also introduces some preliminary knowledge of series. The essential idea is to treat series as partial sum sequences, so that the theory of sequence limits above can be directly applied to obtain convergence tests for series (also divided into general and special two types) and introduces absolute convergence as a stronger form of convergence (unaffected by rearrangement), and the related properties of convergent series (necessary condition - terms $\to 0$). On this basis (mainly the Weierstrass theorem), some useful corollaries are made, such as: the comparison test (only applicable when all terms are non-negative or all terms have the same sign), the Cauchy convergence criterion (based on the relationship between $\sup\lim_{n\to\infty} |a_n|^{1/n}$ and 1) and the d'Alembert test (based on the relationship between $\sup\lim_{n\to\infty} |\frac{a_{n+1}}{a_n}|$ and 1) (applicable in general cases and ensures absolute convergence), and the Cauchy condensation test ($\sum_{n=1}^{\infty} a_n$ and $\sum_{k=0}^{\infty} 2^k a_{2^k}$ have the same convergence behavior) (applicable when terms are non-negative and decreasing).

Examples

1. For the series $\sum_{n=1}^{\infty}(2+(-1)^n)^n x^n$, it converges for $|x| < \frac{1}{3}$ and diverges for $|x| > \frac{1}{3}$ and $|x| = \frac{1}{3}$, using the Cauchy root test.

2. For the series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$, it converges for all $x \in \mathbb{R}$, using the d'Alembert ratio test. Note: This can also be judged by the Cauchy root test, based on $\lim_{n\to+\infty} \sqrt[n]{n!} = +\infty$.

3. For the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, it converges for $p > 1$ and diverges for $p \leq 1$, using the Cauchy condensation test.

3.1.2 ξ.2 The Limit of a Function

The study of function limits largely follows the methods of sequence limits. First, the ε-δ definition of a function limit is introduced (note that function limits are defined on a punctured neighborhood), along with the properties of a function's limit existing at a point (existence $\implies$ local boundedness). On this basis, the four arithmetic operations of function limits are studied (the proof process will become much more convenient if an infinitesimal quantity is first introduced as an equivalent definition of the limit of function) and the inequality relations during the limit process (note: the strictness of inequalities may not be maintained after the limit process). Furthermore, the limit behavior of elementary functions and the limit behavior of composite functions are explained (here, it is necessary to learn two important limits).

A novel aspect of Zorich's approach is his definition of something called a "base" (essentially, a family of sets satisfying certain conditions, which facilitates the description of "approaching" in the limit process); he then re-describes the aforementioned content based on this concept. Next, the discussion proceeds to the problem of the existence of limits (again introducing two methods: the /general/ Cauchy convergence criterion and the /special - monotonic function/ Weierstrass theorem). Finally, the issue of asymptotic comparison of functions (order estimation) is briefly introduced.

2 Important Limits

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1$.
2. $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e$.

3.2 Problems Sets

3.2.1 ξ.1 The Limit of a Sequence

Problem 1. (C3S1Q3)
We mark all the points on a circle obtained from a fixed point by rotations of the circle through angles of n radians, where $n \in \mathbb{Z}$ ranges over all integers. Describe all the limit points of the set so constructed.

Solution. Attempt 1. $n(mod\ \alpha) = n - \alpha \left\lfloor \frac{n}{\alpha} \right\rfloor$, $\forall \alpha \in \mathbb{R}$.
Attempt 2. $\forall \varepsilon > 0$, by pigeonhole principle ($\forall N > 0, as\ n \to \infty$, $n - N$ of integers ($n - N \to \infty$) will be located between those N integers.) $\exists m, n < \frac{1}{\varepsilon}$, s.t. $|m - n|(mod\ 2\pi) < \varepsilon$,
then let $k = |m - n|$, (construct full proof for dense property), $\forall x \in [0, 2\pi]$, $x \in \left[k \left\lfloor \frac{x}{k(mod\ 2\pi)} \right\rfloor, k\lceil \frac{x}{k(mod\ 2\pi)} \rceil\right]$ and the length of interval equals to $k < \varepsilon$, which means $\mathbb{R}_{2\pi}$ is dense in $(0, 2\pi)$ or $[0, 2\pi]$.

Development:
Kronecker's approximation theorem.(Wolfram MathWorld, Wikipedia)

(Variants)
Prove: $\{n^p \pmod{\pi} : p > 0\}$ is dense on $[0, \pi]$.

Proof. Core: Similar to the idea of differential (tool: difference), reduce the order, and finally turn it into the prototype of the above problem.
Let $p > 0$ be an arbitrary real number. We claim that $\{n^p \pmod{\pi} : p > 0\}$ is dense on $[0, \pi]$.
For any function $f(n)$ let $\Delta f(n) = f(n+1) - f(n)$. If $f(n)$ converges to 0 $\pmod{\pi}$ then so does $\Delta f(n)$. Now for $f(n) = n^p$ we have $\Delta f(n) \approx pn^{p-1}$, $\Delta^2 f(n) \approx p(p-1)n^{p-2}$, etc. More precisely, by Newton's binomial theorem, $(n+1)^p - n^p = \sum_{i=1}^{\infty} \binom{p}{i} n^{p-i}$.
Let $k = \lfloor p \rfloor$. One can check that the part of the series with $i > k$ has the overall form $o(1)$. Thus $\Delta(n^p) = pn^{p-1} + c_2 n^{p-2} + \cdots + c_k n^{p-k} + o(1)$ for various constant coefficients $c_2, \ldots, c_k$. Iterating, it follows that $\Delta^k(n^p) = (p)_k n^{p-k} + o(1)$, where $(p)_k = p(p-1)\cdots(p-k+1)$. Thus the function $g(n) = (p)_k n^{p-k}$ tends to 0 mod π, but this is impossible: since π is irrational and we must have $p > k$, but then $g(n) \to \infty$ while $\Delta g(n) \to 0$ (There aren't $\pmod{\pi}$), which means $g(n)$ tends to ∞ "densely".), so the range of g is dense mod π.

Problem 2. (C3S1Q5)

a) $1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n} = 3 - \frac{1}{1 \cdot 2 \cdot 2!} - \cdots - \frac{1}{(n-1) \cdot n \cdot n!}$ holds, as $n \geq 2$.

b) $e = 3 - \sum_{n=0}^{\infty} \frac{1}{(n+1) \cdot (n+2) \cdot (n+2)!}$.

c) For computing e,$e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n}$ is much better than $e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$.

Quick Solution. a) $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}+\frac{1}{1\cdot2\cdot2!}+\cdots+\frac{1}{(n-1)\cdot n\cdot n!}$ $=\left[\frac{1}{n!}+\frac{1}{n!n}+\frac{1}{(n-1)n\cdot n!}-\frac{1}{(n-1)\cdot(n-1)!}\right]=1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{(n-1)!}+\frac{1}{(n-1)\cdot(n-1)!}+\frac{1}{1\cdot2\cdot2!}+\cdots+\frac{1}{(n-2)\cdot(n-1)\cdot(n-1)!}=\cdots=1+\frac{1}{1!}+\frac{1}{1\cdot1!}=3$

b) $\lim_{n\to\infty}1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}$ and $\lim_{n\to\infty}1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}$, $\Delta_n=\frac{1}{n!n}=o(1),n\to\infty$.

c) Since both $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}$ and $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}$ are monotonically increasing $\to e$, and the former has an extra term $\frac{1}{n!n}$ thus it grows faster (i.e., converges faster). To estimate the difference in convergence speed, we just need to find the maximum $k\in\mathbb{N}^+$ such that $\frac{1}{n!n}\geq\frac{1}{(n+1)!}+\cdots+\frac{1}{(n+k)!}$.
(Actually, $\forall n\geq2,k\in\mathbb{N}^+$ i.e $Maxk=+\infty$)

Problem 3. (C3S1Q6)
Given$S_p(a,b)=\left(\frac{a^p+b^p}{2}\right)^{\frac{1}{p}}$ $,p\neq0,a,b>0$, prove:

a) $\forall p\neq0,a,b>0,S_p(a,b)\in[a,b]$;

b) $\lim_{n\to\infty}S_n(a,b)=b,\lim_{n\to\infty}S_{-n}(a,b)=a$.

Quick Solution. a) It is sufficient to define the order relation of a and b, for example: without loss of generality, assume $a<b$.

b) It is sufficient to define the relative size relationship of a and b (i.e., what is commonly called substitution or elimination by increment), for example: without loss of generality, let $b=\lambda a$.

Thoughts:
(Trick) When a problem has multiple (≥ 2) variables, it may be helpful to look for connections between these seemingly "independent" variables (for example, in question 6.b) of this section).

Problem 4. (C3S1Q7) (Classical problem)
If $a>0$, then sequence $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)$ converges to $\sqrt{a}$.

Quick Solution. $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)\geq\sqrt{a},\ n\geq1$, i.e. $\forall x_1>0,\ x_2\geq\sqrt{a}$.
[Induction] then $\{x_n\}_{n=2}^{\infty}$ monotonically decrease, $\lim_{n\to\infty}x_{n+1}=\lim_{n\to\infty}\frac{1}{2}\left(x_n+\frac{1}{x_a}\right)=\sqrt{a}$.

Thoughts:
(Experience) An experienced student, upon seeing a recurrence relation, can immediately guess that the problem will most likely require the use of methods like the Weierstrass theorem to first determine convergence, and then take the limit on both sides of the equation.

(Tip) If starting from a general problem-solving approach, the first step is usually to determine the basic properties of the sequence (positivity/negativity, monotonicity, parity, periodicity (or these properties starting from a certain term), etc.), and then consider which method to use to address convergence (general $\to$ Cauchy convergence criterion, monotonic $\to$ Weierstrass theorem) and other issues.

(Follow-up Question)
Please estimate the convergence rate, i.e., the dependence of $x_n - \sqrt{a}$ on n.

Problem 5. (C3S1Q8)

a) (Faulhaber's formula) $S_k(n) = \sum_{i=1}^{n} i^k = \frac{1}{k+1}\sum_{r=0}^{k} C_{k+1}^r B_r n^{k+1-r}$, B_r is Bernoulli number.

b) $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \frac{1}{k+1}$.

Solution. a) This part is a little hard, because to wholy solving it, we needs expotential generating function with indeterminate z: $G(z,n) = \sum_{k=0}^{\infty} S_k(n)\frac{1}{k!}z^k$. You could get it from wikipedia.

b) Method1. $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \frac{i^k}{n^k} = \int_0^1 x^k dx = \frac{1}{k+1}$ (Riemann sum)

Method2. $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \lim_{n\to\infty} \frac{(n+1)^k}{(n+1)^{k+1}-n^{k+1}}$ (Stolze-Cesaro Lemma) $= \lim_{n\to\infty} \frac{1}{(n+1)-\left(\frac{n}{n+1}\right)^k\cdot n} = \lim_{n\to\infty} \frac{1}{1+n\left[1-\left(\frac{n}{n+1}\right)^k\right]}$ (order estimation) $= \lim_{n\to\infty} \frac{1}{1+n\left[1-\left(1-k\cdot\frac{1}{n+1}+O\left(\frac{1}{n^2}\right)\right)\right]} = \lim_{n\to\infty} \frac{1}{1+\frac{n}{n+1}\cdot k+O\left(\frac{1}{n}\right)} = \frac{1}{1+k}$.

3.2.2 ξ.2 The Limit of a Function

Problem 6. (C3S2Q1)
Porve that:

a) there $\exists$ a unique function defined on $\mathbb{R}$ satisfying : $f(1) = a$ (where $a > 0, a \neq 1$), $f(x_1)\cdot f(x_2) = f(x_1+x_2)$, $\lim_{x\to x_0} f(x) = f(x_0)$.

b) there $\exists$ a unique function defined on $\mathbb{R}^+$ satisfying : $f(a) = 1$ (where $a > 0, a \neq 1$), $f(x_1)+f(x_2) = f(x_1\cdot x_2)$, $\lim_{x\to x_0} f(x) = f(x_0)$.

Quick Solution. By substitution (for example: $x \triangleq e^{\xi}$, etc.), transform the given variant of the Cauchy equation back into the form $f(x+y) = f(x)+f(y)$ (uniqueness is obtained from the equation itself and continuity, and the method has been provided in the discussion of the corresponding equation in Fikhtengoltz's "Calculus Tutorial (I)").

Problem 7. (C3S2Q2)

a) Provide an isomorphism mapping between the group $(\mathbb{R}, +)$ and the group $(\mathbb{R}^+, \cdot)$;

b) Prove that $(\mathbb{R}, +)$ and the group $(\mathbb{R} \setminus 0, \cdot)$ are not isomorphic.

Quick Solution. a) $\forall x, y \in \mathbb{R}$, if $\varphi(x+y) = \varphi(x) \cdot \varphi(y)$, then $\varphi(x) = e^{cx}$;

b) From the fact that "isomorphism not only is a bijection but also preserves the group operation", if $\varphi : \mathbb{R} \to \mathbb{R}\backslash 0$ exists, then from $\varphi(0) = 1$, it follows that $\varphi(x) \cdot \varphi(-x) = \varphi(0) = 1$.

Problem 8. (C3S2Q3)

Quick Solution. a) $\lim\limits_{x\to+0} x^x = \lim\limits_{x\to+0} e^{x\ln(x)} = \lim\limits_{t\to+\infty} e^{-\frac{\ln(t)}{t}} = 1$;

b) $\lim\limits_{x\to+\infty} x^{\frac{1}{x}} = \lim\limits_{x\to+\infty} e^{\frac{\ln(x)}{x}} = 1$;

c) $\lim\limits_{x\to 0} \frac{\log_a(1+x)}{x} = \lim\limits_{x\to 0} \frac{1}{\ln a} \cdot \frac{\ln(1+x)}{x} = \left[\ln(1+x) = x - \frac{1}{2}x^2 + O(x^3)\right] = \frac{1}{\ln a}$;

d) $\lim\limits_{x\to 0} \frac{a^x-1}{x} = [a^x - 1 \triangleq t] = \lim\limits_{x\to 0} \frac{t}{\log_a(1+t)} = \ln a$.

Notes. Both a) and b) used $\lim_{x\to+\infty} \frac{\log_a x}{x^\alpha} = 0,\ \alpha > 0$.

Problem 9. (C3S2Q4)
Prove: $1 + \frac{1}{2} + \cdots + \frac{1}{n} = \ln n + c + o(1),\ n \to \infty$.

Quick Solution. Using $\ln(n+1) - \ln n = \ln \frac{n+1}{n} = \ln\left(1 + \frac{1}{n}\right) = \frac{1}{n} + O\left(\frac{1}{n^2}\right),\ n \to \infty$, then Σ them.

Problem 10. (C3S2Q5)

a) If two positive term series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ satisfy: $a_n \sim b_n, n \to \infty$, then these two series have the same convergence behavior;

b) The series $\sum_{n=1}^{\infty} \sin \frac{1}{n^p}$ converges only when $p > 1$.

Solution. a) $a_n \sim b_n$ i.e. $\forall \varepsilon > 0, \exists N > 0, s.t. \forall n > N, |\frac{a_n}{b_n} - 1| < \varepsilon$ i.e. $1-\varepsilon < \frac{a_n}{b_n} < 1+\varepsilon, \forall n > N$, take a specific $\varepsilon = \varepsilon_0$, then $(1-\varepsilon_0)b_n < a_n < (1+\varepsilon_0)b_n$. Convergence/divergence is thereby controlled.

b) $p > 0 : \sin\frac{1}{n^p} \sim \frac{1}{n^p}$ Two positive term series have the same convergence behavior, the convergence/divergence of the latter can be given by the Cauchy condensation test (see Summary C3S1 for details)
$p = 0 : \sin\frac{1}{n^p} = \sin 1 > 0$ diverges
$p < 0 : \sum_{n=1}^{\infty} \sin\frac{1}{n^p} = [q \triangleq -p > 0] = \sum_{n=1}^{\infty} \sin n^q$ diverges (Essentially, $\{n^q (\bmod \pi)\}$ is dense on $[0, \pi]$, details in (C3S1Q3 variants).

Problem 11. (C3S2Q6)

a) If $\forall n \in \mathbb{N}$ we have $a_n \geq a_{n+1} > 0$, and the series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n = o\left(\frac{1}{n}\right)$, $n \to \infty$;

b) If $b_n = o\left(\frac{1}{n}\right)$, then it is always possible to construct a convergent series $\sum_{n=1}^{\infty} a_n$; such that $b_n = o(a_n)$, $n \to \infty$;

c) If the positive term series $\sum_{n=1}^{\infty} a_n$ converges, then the series $\sum_{n=1}^{\infty} A_n$, as $A_n = \sqrt{\sum_{k=n}^{\infty} a_k} - \sqrt{\sum_{k=n+1}^{\infty} a_k}$, also converges, and $a_n = o(A_n), n \to \infty$;

d) If the positive term series $\sum_{n=1}^{\infty} a_n$ diverges, then the series $\sum_{n=1}^{\infty} A_n$, as $A_n = \sqrt{\sum_{k=1}^{n} a_k} - \sqrt{\sum_{k=1}^{n-1} a_k}$, also diverges, and $A_n = o(a_n), n \to \infty$.

Solution. a) Applying the Cauchy convergence criterion to the series $\sum_{n=1}^{\infty} a_n$, we have: $\forall \varepsilon > 0, \exists N > 0$ such that $\forall n, m > N, |a_n + \cdots + a_m| < \varepsilon$. If we set $m = 2n$ and since a_n is monotonically decreasing, then $na_{2n} < \varepsilon$, i.e. $a_n = o\left(\frac{1}{n}\right)$;

b) Consider alternating series and monotonic sequences: $a_n \triangleq (-1)^n \sup_{k \geq n} b_k$, then $\sum_{n=1}^{\infty} a_n$ converges. (Leibniz Theorem) Or just let $a_n = (-1)^n \frac{1}{n}$.

c) d) Easy to prove. (Cancellation of intermediate terms in summation; rationalization, division).

Thoughts:
From c) and d), it can be seen: there always exists a convergent positive term series that is "larger" than a given convergent positive term series, and there always exists a positive term series that is "smaller" than a given divergent positive term series.
There is no unique, universal standard series that can be applied to all comparison tests.

Problem 12. (C3S2Q7&Q8)

a) The series $\sum_{n=1}^{\infty} \ln a_n, a_n > 0, n \in \mathbb{N}$ converges i.f.f. the sequence $\{\Pi_n = a_1 \cdots a_n\}$ has a nonzero limit;

b) If the infinite product $\prod_{n=1}^{\infty} e_n$ converges, then $\lim_{n\to\infty} e_n = 1$;

c) The series $\sum_{n=1}^{\infty} \ln(1+a_n), |a_n| < 1$ converges absolutely i.f.f. the series $\sum_{n=1}^{\infty} a_n$ converges absolutely;

d) The infinite product $\prod_{n=1}^{\infty} e_n, e_n > 0$ converges i.f.f. the series $\sum_{n=1}^{\infty} \ln e_n$ converges;

e) If $e_n = 1 + a_n$ and a_n all have the same sign, then the infinite product $\prod_{n=1}^{\infty}(1+a_n)$ converges i.f.f. the series $\sum_{n=1}^{\infty} a_n$ converges.

Quick Solution. a) Easy to prove. $\sum_{n=1}^{\infty} \ln a_n = \ln \prod_{n=1}^{\infty} a_n$ (they have the same convergence behavior).

b) Easy to prove. (If $\prod_{n=1}^{\infty} e_n = c \neq 0$, then $\lim_{n\to\infty} e_n = \lim_{n\to\infty} \frac{\prod_{k=1}^{n} e_k}{\prod_{k=1}^{n-1} e_k} = 1$. Alternatively, take logarithms and apply the basic properties of series convergence.)

c) If $\sum_{n=1}^{\infty} \ln(1+a_n)$ converges, then by a) $\prod_{n=1}^{\infty}(1+a_n)$ has a nonzero limit. Further, by b), $a_n \to 0$ as $n \to \infty$, and since $|\ln(1+a_n)| \sim \ln(1+|a_n|) \sim |a_n|$, it follows (from C3S2Q5 a)) that the statement is proved.

d) Easy to prove. (Same as a))

e) $\ln(1+a_n) \sim a_n$, so $\prod_{n=1}^{\infty}(1+a_n)$ converges if and only if $\sum_{n=1}^{\infty} \ln(1+a_n)$ converges if and only if $\sum_{n=1}^{\infty} a_n$ converges.

Problem 13. (C3S2Q9)

a) Calculate $\Pi_{n=1}^{\infty}(1+x^{2^{n-1}})$;

b) Caculate $\Pi_{n=1}^{\infty} \cos\left(\frac{x}{2^n}\right)$, and prove $\frac{\pi}{2} = \frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}}}}\cdots}$ (Vieta's formula);

c) Find the $f(x)$, s.t. $f(0) = 1,\ f(2x) = \cos^2 x \cdot f(x),\ \lim_{x\to 0} f(x) = f(0)$.

Solution. a) From C3S2Q7 e), $\prod_{n=1}^{\infty}(1+x^{2^n-1})$ and $\sum_{n=1}^{\infty} x^{2^n-1}$ have the same convergence behavior, i.e., they converge only when $|x| < 1$, and $\prod_{n=1}^{\infty}(1+x^{2^n-1}) = \frac{1-x^{2^n}}{1-x}$ (generalized power series multiplication theorem);

b) $\prod_{n=1}^{\infty} \cos\left(\frac{x}{2^n}\right) = \lim_{n\to\infty} \prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) = \lim_{n\to\infty} \frac{\sin\frac{x}{2^n}\prod_{k=1}^{n}\cos\frac{x}{2^k}}{\sin\frac{x}{2^k}} = \lim_{n\to\infty} \frac{\sin x}{2^n \sin\frac{x}{2^n}} = \lim_{n\to\infty} \frac{\sin x}{x} \cdot \frac{\frac{x}{2^n}}{\sin\frac{x}{2^n}} = \frac{\sin x}{x}$ Due to $\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$ and (above) $\prod_{n=1}^{\infty} \cos\frac{x}{2^n} = \frac{\sin x}{x}$ (let $x = \frac{\pi}{2}$), Vieta's formula is proved;

c) Using the recurrence relation $f(2x) = \cos^2 x \cdot f(x)$ and taking $\frac{x}{2^n}, n = 0, 1, 2, \ldots$ yields, $f(x) = \left(\frac{\sin x}{x}\right)^2$.

Problem 14. (C3S2Q10)

a) If $\frac{b_n}{b_{n+1}} = 1+\beta_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \beta_n$ converges absolutely, then the limit $\lim_{n\to\infty} b_n = b$ evidently exists;

b) If $\frac{a_n}{a_{n+1}} = 1 + \frac{p}{n} + \alpha_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \alpha_n$ converges absolutely, then $a_n \sim \frac{c}{n^p}, n \to \infty$;

c) If $\frac{a_n}{a_{n+1}} = 1 + \frac{p}{n} + \alpha_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \alpha_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n$ converges when $p > 1$ and diverges when $p \le 1$ (Gauss's test for absolute convergence of series).

Solution. a) From C3S2Q7 and Q8 c), it follows that $\sum_{n=1}^{\infty} \ln(1 + \beta_n) = \sum_{n=1}^{\infty} \ln\left(\frac{b_n}{b_{n+1}}\right)$ and $\sum_{n=1}^{\infty} \beta_n$ have the same convergence behavior, both absolutely convergent.

b) (Good question, good solution)
Method 1. Consider $\frac{n^p}{(n+1)^p}\frac{a_n}{a_{n+1}} = \frac{n^p}{(n+1)^p}\left(1 + \frac{p}{n} + \alpha_n\right) = \left(1 + \frac{1}{n}\right)^{-p} + \left(1 + \frac{1}{n}\right)^{-p}\frac{p}{n} + \left(1 + \frac{1}{n}\alpha_n\right)^{-p} = \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\left(1 + \frac{p}{n}\right) + \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\alpha_n \left[\triangleq 1 + \theta_n\right]$,
$\theta_n = \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\left(1 + \frac{p}{n} + \alpha_n\right) - 1 = O\left(\frac{1}{n^2}\right)\left(1 + \frac{p}{n} + \alpha_n\right) - \frac{p^2}{n^2} + \alpha_n\left(1 - \frac{p}{n}\right)$, $|\theta_n| \le |O\left(\frac{1}{n^2}\right)\left(1 + \frac{p}{n} + \alpha_n\right)| + |\frac{p^2}{n^2}| + |\alpha_n\left(1 - \frac{p}{n}\right)|$, and $\sum_{n=1}^{\infty} \alpha_n$ abs. cgt. , then $\sum_{n=1}^{\infty} \theta_n$ abs. cgt. Then due to C3S2Q7 c), $\sum_{n=1}^{\infty} \ln(1 + \theta_n) = \sum_{n=1}^{\infty} \ln\frac{n^p a_n}{(n+1)^p a_{n+1}} = \ln\alpha_1 - \lim_{n\to\infty} n^p\alpha_n$ abs. cgt. i.e. $n^p\alpha_n \sim c, n \to \infty$. $(\alpha_n \sim \frac{c}{n^p})$.
Method 2 (Optimize). Following the first step of Method 1:
$\ln\frac{n^p a_n}{(n+1)^p a_{n+1}} = \ln\frac{\alpha_n}{\alpha_{n+1}} - p\ln\left(1 + \frac{1}{n}\right) = \ln\left(1 + \frac{p}{n} + \alpha_n\right) - p\ln\left(1 + \frac{1}{n}\right)$
$= \frac{p}{n} + \alpha_n + O\left(\frac{p}{n} + \alpha_n\right)^2 - p\left(\frac{1}{n} + O\left(\frac{1}{n^2}\right)\right) = \alpha_n + O\left(\frac{1}{n^2}\right) + O(\alpha_n) \to$ abs. cgt. going on Method 1.

c) From b) and the convergence behavior of $\sum_{n=1}^{\infty}\frac{1}{n^p}$, it immediately follows.

Problem 15. (C3S2Q11)

Prove: For any positive-term sequence $\{a_n\}$, $\limsup_{n\to\infty}\left(\frac{1+a_{n+1}}{a_n}\right)^n \ge e$, and this estimate cannot be improved.

Solution. (Great deal: Wonderful observation!) /Contradiction/
Lemma: $\forall x \in (0, e)$, for sufficiently large n, $x^{\frac{1}{n}} < 1 + \frac{1}{n}$.
If $\overline{\lim_{n\to\infty}}\left(\frac{1+a_{n+1}}{a_n}\right)^n < e$, $a_n > 0$, then $\exists b$, s.t. $\overline{\lim_{n\to\infty}}\left(\frac{1+a_{n+1}}{a_n}\right)^n < b < e$, i.e. $\exists$ sufficient large n, s.t. $\left(\frac{1+a_{n+1}}{a_n}\right)^n < b < e$, i.e. (using std::Lemma) $\frac{1+a_{n+1}}{a_n} < b^{\frac{1}{n}} < \frac{n+1}{n}$, let $b_n \triangleq \frac{a_n}{n}$, then $b_{n+1} < b_n - \frac{1}{n+1}$, increasing to $n+m$ and Σ them: $b_{n+m} < b_n - \sum_{k=1}^{m}\frac{1}{n+k}$, fixing b_n and letting $m \to \infty$ we have $b_{n+m} = \frac{a_{n+m}}{n} \to -\infty,\ m \to \infty$ which derivate constraction.

4 Basel Problem

4.1 $\zeta(2) = \sum\limits_{n=1}^{\infty} \frac{1}{n^2}$

Problem 16. Calculate (or prove)

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Solution. Method 1. Double Integral.
Consider the double integral $I = \iint_{D_1} \frac{1}{1-xy}\, dx\, dy$, where $D_1 = \{(x, y) : 0 < x < 1, 0 < y < 1\}$.

$$\begin{aligned} \iint_{D_1} \frac{dx\, dy}{1-xy} &= \text{/Generalized power series multiplication theorem/} \\ &= \iint_{D_1} \sum_{n=0}^{\infty} (xy)^n\, dx\, dy = \int_0^1 \left(\int_0^1 \sum_{n=0}^{\infty} (xy)^n\, dx \right) dy \\ &= \int_0^1 \sum_{n=1}^{\infty} \frac{y^{n-1}}{n}\, dy = \sum_{n=1}^{\infty} \frac{1}{n^2}. \end{aligned}$$

/Transformation of coordinates or variable substitution/

$$\begin{aligned} &[x = \frac{u-v}{\sqrt{2}}, y = \frac{u+v}{\sqrt{2}}, \frac{D(x, y)}{D(u, v)} = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} \\ &\Leftarrow (u, v)^T = \begin{pmatrix} \cos -\frac{\pi}{4} & -\sin -\frac{\pi}{4} \\ \sin -\frac{\pi}{4} & \cos -\frac{\pi}{4} \end{pmatrix} (x, y)^T] \end{aligned}$$

$$\begin{aligned} I &= \iint_{D_2} \frac{1}{1 - \frac{u^2-v^2}{2}} \cdot 1\, du\, dv \\ &= 2 \left[\iint_{D_21} \frac{1}{2 - u^2 + v^2}\, du\, dv + \iint_{D_22} \frac{1}{2 - u^2 + v^2}\, du\, dv \right] \\ &= 4 \left[\int_0^{\frac{\sqrt{2}}{2}} \left(\int_0^u \frac{dv}{2 - u^2 + v^2} \right) du + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \left(\int_0^{\sqrt{2}-u} \frac{dv}{2 - u^2 + v^2} \right) du \right] \\ &= [u \triangleq \sqrt{2} \sin\theta, v \triangleq \sqrt{2} \cos\theta] \\ &= \frac{\pi^2}{18} + \frac{\pi^2}{9} = \frac{\pi^2}{6}. \end{aligned}$$

i.e. $\zeta(2) = \sum\limits_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

5 Chapter 4.Continuous Functions

5.1 Comprehensions

This chapter discusses the continuity of functions (including discontinuities when not continuous). First, we define the concepts of continuity at a point, continuity on an interval, and one-sided continuity, and introduces the concept of points of discontinuity (2 types: depending on whether the one-sided limits exist). Further, it defines the stronger concept of uniform continuity. Then, it presents the arithmetic operations and composition of continuous functions, and provides 3 global properties of continuous functions (on closed intervals: 1. Bolzano-Cauchy intermediate value theorem, 2. Weierstrass extreme value theorem (boundedness), 3. Cantor-Heine uniform continuity theorem). Additionally, Zorich also covers the continuity of monotonic functions in the book (which is special as before) (types of discontinuities for monotonic functions (only the first kind) and their number (at most countable), and the relationship between continuity and the range (intermediate value property) for monotonic functions derived from this). Finally, it introduces the inverse function theorem, which is very intuitively understood (on closed intervals: existence $\Leftrightarrow$ injective $\Leftrightarrow$ strictly monotonic, preserving continuity).

Continuity of 2 Special fuctions:

1. Dirichlet function $\mathcal{D}(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$

   Equivalent expression: $\forall x \in \mathbb{R}, \mathcal{D}(x) = \lim_{k\to\infty}(\lim_{j\to\infty}(\cos(k!x))^{2j})$
   Properties: Every point is a discontinuity point of the second kind (no limit anywhere, discontinuous, non-differentiable); has every positive rational number as its period (thus no minimal positive period); not Riemann integrable on any bounded interval, but Lebesgue integrable on $[0, 1]$.

2. Riemann function $\mathcal{R}(x) = \begin{cases} \frac{1}{n}, & x = \frac{m}{n} \in \mathbb{Q} \setminus \{0\} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \cup \{0\} \end{cases}$

   The number-theoretic essence of the first kind discontinuity points of the Riemann function: $\forall a \in \mathbb{R}, N \in \mathbb{N}$, in $U(a)$ there are only finitely many rational numbers m/n (assumed to be in irreducible form) with $n < N$, i.e., $\frac{1}{n} > \frac{1}{N}$.
   Properties: $\mathcal{R}(x)$ is continuous at all irrational points and has first kind discontinuities at all rational points; the Riemann function is Riemann integrable.

5.2 Problems Sets

Problem 17. (C4S2Q1)

Quick Solution. a) $f \in C(A), B \subset A \implies f|_B \in C(B)$.

b) $f : E_1 \cup E_2 \to \mathbb{R}$, satisfying $f|_{E_i} \in C(E_i), i = 1, 2 \not\Rightarrow f \in C(E_1 \cup E_2)$, counterexample: $f(x) = \tan x$.

c) $\forall N > 0, \{\frac{m}{n} : n < N, (m, n) = 1, m, n \in \mathbb{Z}^+\}$ is a finite set, i.e. $\exists U(x; \delta)(\forall x \in (0, 1))$ such that $\forall x_1, x_2 \in U(x; \delta), |R(x_1) - R(x_2)| \leq \frac{1}{N} < \varepsilon$ (Cauchy convergence criterion), then $\forall x_0 \in [0, 1], \lim_{x\to x_0} R(x) = 0$. (The proof method is consistent with the discussion of discontinuity points of the Riemann function in Example 2 of C4 Summary.)

Problem 18. (C4S2Q2)
If $f \in C[a, b]$, then $m(x) = \min_{a\leq t\leq x} f(t), \max_{a\leq t\leq x} f(t) \in C[a, b]$.

Quick Solution. Note that $m(x) = \min_{a\leq t\leq x} f(t)$, $M(x) = \max_{a\leq t\leq x} f(t) \in C[a, b]$, and since each is monotonic, their ranges can be given. Then, by relating to the relationship between the continuity of monotonic functions and their ranges (intermediate value property), the proof is completed.

Problem 19. (C4S2Q3)

a) On an open interval, the inverse function of a monotonic function exists and is continuous.

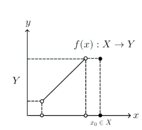
b) Construct a monotonic function with a countable set of discontinuities.

c) If $f : X \to Y$ and $f^{-1} : Y \to X$ are inverse functions of each other, and f is continuous at $x_0 \in X$, then it does not imply that f^{-1} is continuous at $y_0 = f(x_0)$.

Solution. a) Extend to the closed interval (taking one-sided limits at the endpoints) using the inverse function theorem, then restrict back to the open interval.

b) $f(x) = [x]$.

c) Actually, such a counterexample comes from the lack of closed interval restriction in the inverse function theorem. If x_0 is an isolated point, it is easy to construct an example.



Problem 20. (C4S2Q4)

a) if $f \in C[a,b]$ and $g \in C[a,b]$, and, in addition, $f(a) < g(a)$ and $f(b) > g(b)$, then there exists a point $c \in [a,b]$ at which $f(c) = g(c)$;

b) any continuous mapping $f : [0,1] \to [0,1]$ of a closed interval into itself has a fixed point, that is, a point $x \in [0,1]$ such that $f(x) = x$;

c) if two continuous mappings f and g of an interval into itself commute, that is, $f \circ g = g \circ f$, then they have a common fixed point;

d) a continuous mapping $f : \mathbb{R} \to \mathbb{R}$ may fail to have a fixed point;

e) a continuous mapping $f :]0,1[\to]0,1[$ may fail to have a fixed point;

f) if a mapping $f : [0,1] \to [0,1]$ is continuous, $f(0) = 0$, $f(1) = 1$, and $(f \circ f)(x) \equiv x$ on $[0,1]$, then $f(x) \equiv x$;

Quick Solution. a) Constrcut $h(x) = f(x) - g(x)$, then Bolzano-Cauchy intermediate value theorem applies.

b) Constrcut $h(x) = f(x) - x$, then disccuss its values at $x = 0$ and $x = 1$, and using Bolzano-Cauchy IVT.

c) /Insane/ The question should be modified to 'If 2 continuous maps f and g of a line segment into itself commute, i.e. $f \circ g = g \circ f$, then they donn't always have a common fixed point, though for linear maps and polynomials in general we have such a point.'
/The counterexample was given by W. M. Boyce 1969, the proof for linear maps is easy using [contradiction], the proof for polynomials is a bit of hard (wait to add)./

d) $f(x) = x + 1$, $f(x) = e^x$

e) $f(x) = \sin \frac{\pi}{2} x$, $f(x) = x^2$

f) 1°. f is injection [contradiction] and surjection $\Longrightarrow$ bijection, then we could get monotonic; 2°. $f \equiv x \Leftrightarrow f \leq x \wedge f \geq x$ [contradiction].

$$\begin{tikzcd} x \arrow[r, bend left, "f"] & f(x) \arrow[l, bend left, "f"] \end{tikzcd}$$

Problem 21. (C4S2Q5)
Show that the set of values of any function that is continuous on a closed interval is a closed interval.

Quick Solution. Weierstrass exreme value theorem + Bolzano-Cauchy intermediate value theorem.

Problem 22. (C4S2Q6)
Prove the following statements:

a) If a mapping $f : [0,1] \to [0,1]$ is continuous, $f(0) = 0$, $f(1) = 1$, and $f^n(x) := \underbrace{f \circ \ldots \circ f}_{n \text{ factors}}(x) \equiv x$ on $[0,1]$, then $f(x) \equiv x$.

b) If a function $f : [0,1] \to [0,1]$ is continuous and nondecreasing, then $\forall x \in [0,1]$ at least one of the following situations must occur: either x is a fixed point, or $f^n(x)$ tends to a fixed point. (Here $f^n(x) = f \circ \ldots \circ f(x)$ is the nth iteration of f.)

Solution. a) Similiar (almost same) to C4S2Q4-f).

b) $\forall x \in [0,1]$, $x_0 = x, x_1 = f(x), \cdots, x_n = f^n(x), \cdots$,
then there are only 2 condition between x and $f(x)$:
1. $x = f(x)$ i.e. fixed point;
2. $x \leq f(x)$ or $x \leq f(x)$, from monotonicity of $f(x)$ we know $\{x_n\}_{n=0}^{\infty}$ monotonic and bounded, then due to continuity of $f(x)$ we could find the limit (fixed point).

Problem 23. (C4S2Q7)
Let $f : [0,1] \to \mathbb{R}$ be a continuous function such that $f(0) = f(1)$. Show that

a) $\forall n \in \mathbb{N}$ there $\exists$ a horizontal closed interval of length $\frac{1}{n}$ with endpoints on the graph of this function;

b) if the number l is not of the form $\frac{1}{n}$ there exists a function of this form on whose graph one cannot inscribe a horizontal chord of length l.

Solution. a) It $\Leftrightarrow \forall n \in \mathbb{N}, \exists x \in [0, 1 - \frac{1}{n}]$, s.t. $f(x + \frac{1}{n}) - f(x) = 0$.
Construct $g(x) = f(x + \frac{1}{n}) - f(x)$, $x \in [0, 1 - \frac{1}{n}]$.
Consider a partition of $[0, 1 - \frac{1}{n}]$ with n points: $0, \frac{1}{n}, \ldots, 1 - \frac{1}{n}$ and their $g(x)$ values:

$$\begin{cases} g(0) = f(\frac{1}{n}) - f(0) \\ g(\frac{1}{n}) = f(\frac{2}{n}) - f(\frac{1}{n}) \\ \cdots \\ g(1 - \frac{1}{n}) = f(1) - f(1 - \frac{1}{n}) \end{cases}, \ \Sigma\text{: } \sum_{k=0}^{n-1} g(\tfrac{k}{n}) = f(1) - f(0) \equiv 0$$

then $g(0), g(\frac{1}{n}), \ldots, g(1 - \frac{1}{n})$ either all 0, or have at least 2 different signs (then using Bolzano-Cauchy IVT), thus there always $\exists x \in [0, 1 - \frac{1}{n}]$, s.t. $g(x) = f(x + \frac{1}{n}) - f(x) = 0$.

b) $\forall l > 0$, consider: $f(x) = \sin^2(\frac{\pi x}{r}) - x \sin^2(\frac{\pi}{r}), \quad x \in [0,1]$
Clearly, $f(0) = f(1)$. If $f(x) = f(x + r)$, then $r \sin^2(\frac{\pi}{r}) = 0$, which means $r = \frac{1}{n}$, $n \in \mathbb{N}$. (by Paul Levy)
/Development: Universal Chord Theorem (by Paul Levy)/

Problem 24. (C4S2Q8)
The *modulus of continuity* of a function $f : E \to \mathbb{R}$ is the function $\omega(\delta)$ defined for $\delta > 0$ as follows: $\omega(\delta) = \sup\limits_{\substack{|x_1-x_2|<\delta \\ x_1,x_2\in E}} |f(x_1) - f(x_2)|$.
Thus, the least upper bound is taken over all pairs of points x_1, x_2 of E whose distance apart is less than δ.
Show that:

a) the modulus of continuity is a nondecreasing nonnegative function having the limit $\omega(+0) = \lim\limits_{\delta\to+0} \omega(\delta)$;

b) for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any points $x_1, x_2 \in E$ the relation $|x_1 - x_2| < \delta$ implies $|f(x_1) - f(x_2)| < \omega(+0) + \varepsilon$;

c) if E is a closed interval, an open interval, or a half-open interval, the relation $\omega(\delta_1 + \delta_2) \leq \omega(\delta_1) + \omega(\delta_2)$ holds for the modulus of continuity of a function $f : E \to \mathbb{R}$;

d) the moduli of continuity of the functions x and $\sin(x^2)$ on the whole real axis are respectively $\omega(\delta) = \delta$ and the constant $\omega(\delta) = 2$ as $\delta > 0$;

e) a function f is uniformly continuous on E if and only if $\omega(+0) = 0$.

Solution. a) Consider set $W(\delta) = \{|f(x_1) - f(x_2)|\colon |x_1 - x_2| < \delta, x_1, x_2 \in E\}$, then $\forall\delta_1 < \delta_2$, $W(\delta_1) \subseteq W(\delta_2)$, then $\omega(\delta_1) = \sup\limits_{\substack{|x_1-x_2|<\delta_1 \\ x_1,x_2\in E}} |f(x_1) - f(x_2)| \leq \sup\limits_{\substack{|x_1-x_2|<\delta_2 \\ x_1,x_2\in E}} |f(x_1) - f(x_2)| = \omega(\delta_2)$.
If $f(x)$ is bounded on E, then $\exists$ finite $\omega(+0) = \lim_{\delta\to+0} \omega(\delta)$.

b) By definition of $\omega(+0) = \lim_{\delta\to+0} \omega(\delta)$, $\forall\varepsilon > 0$, $\exists\delta > 0$ s.t. $|\sup\limits_{\substack{|x_1-x_2|<\delta \\ x_1,x_2\in E}} |f(x_1) - f(x_2)| - \omega(+0)| < \varepsilon$, expand it.

c) Using: if $|x - y| < d_1 + d_2$, then $\exists z$, s.t. $|x - z| < d_1, |y - z| < d_2$.

d) $f(x) = x$: manifestly; $f(x) = \sin(x^2)$: essentially, $\forall\varepsilon > 0$, $\exists k > 0$, s.t. $\sqrt{(2k + 1)\pi} - \sqrt{(2k)\pi} < \varepsilon$.

e) Necessity "$\Rightarrow$": by definition of uniform continuity; Sufficiency "$\Leftarrow$": by contradiction.

Problem 25. (C4S2Q9)
Let f and g be bounded functions defined on the same set X. The quantity $\Delta = \sup\limits_{x\in X} |f(x) - g(x)|$ is called the *distance* between f and g. It shows how well one function approximates the other on the given set X. Let X be a closed interval $[a, b]$. Show that if $f, g \in C[a, b]$, then $\exists x_0 \in [a, b]$, where $\Delta = |f(x_0) - g(x_0)|$, and that such is not the case in general for arbitrary bounded functions.

Quick Solution. 1. $h(x) = |f(x) - g(x)|$ is continuous on $[a, b]$, then by Weierstrass extreme value theorem (sup = max).
2. Counterexample: construct using discontinuities.

Problem 26. (C4S2Q10)
Let $P_n(x)$ be a polynomial of degree n. We are going to approximate a bounded function $f : [a, b] \to \mathbb{R}$ by polynomials. Let

$$\Delta(P_n) = \sup_{x\in[a,b]} |f(x) - P_n(x)| \text{ and } E_n(f) = \inf_{P_n} \Delta(P_n),$$

where the infimum is taken over all polynomials of degree n. A polynomial P_n is called a *polynomial of best approximation* of f if $\Delta(P_n) = E_n(f)$.
Show that:

a) there exists a polynomial $P_0(x) \equiv a_0$ of best approximation of degree zero;

b) among the polynomials $Q_\lambda(x)$ of the form $\lambda P_n(x)$, where P_n is a fixed polynomial, there is a polynomial Q_{λ_0} such that $\Delta(Q_{\lambda_0}) = \min_{\lambda\in\mathbb{R}} \Delta(Q_\lambda)$;

c) if there exists a polynomial of best approximation of degree n, there also exists a polynomial of best approximation of degree $n + 1$;

d) for any bounded function on a closed interval and any $n = 0, 1, 2, \ldots$ there exists a polynomial of best approximation of degree n.

Quick Solution. a) $P_0(x) \equiv a_0 = \frac{\sup f(x)+\inf f(x)}{2}$.

b) $\Delta(Q_\lambda) = \sup_{x\in[a,b]} |\lambda P_n(x) - f(x)|$, then discuss the uniform continuity of $\Delta(Q_\lambda)$ w.r.t. λ.

Lemma. $\sup\limits_{x\in[a,b]} |f-g| \le \sup\limits_{x\in[a,b]} |f-h| + \sup\limits_{x\in[a,b]} |h-g|$, $\forall f, g, h,\ x \in [a, b]$

Proof idea: $|f - g| \le |f - h| + |h - g|$.
Using Lemma, $\forall \lambda_i,\ i = 1, 2,$ $\sup\limits_{x\in[a,b]} |f(x) - \lambda_1 P_n(x)| \le \sup\limits_{x\in[a,b]} |f(x) - \lambda_2 P_n(x)| + \sup\limits_{x\in[a,b]} |\lambda_1 P_n(x) - \lambda_2 P_n(x)|$, i.e. $|\Delta(Q_{\lambda_1}) - \Delta(Q_{\lambda_2})| \le \sup\limits_{x\in[a,b]} |\lambda_1 P_n(x) - \lambda_2 P_n(x)| = |\lambda_1 - \lambda_2| \sup\limits_{x\in[a.b]} |P_n(x)| = \varepsilon\mathcal{O}(1)$.

c) Proof idea: /Great idea/ We exploit the existence of a minimum value for a continuous function on a compact set. $\to$ Specifically, we treat the coefficient space of the $n + 1$ degree polynomial as a finite-dimensional Euclidean space and define a continuous error function on this space. Then restrict the coefficients to a compact set, ensuring that the error function reaches its minimum on this set. This minimum is the $n + 1$ best approximation error.

d) Induction using c).

Remark: Whole proof of c)
Assume there exists a degree-n best approximation polynomial P_n^* such that $\Delta(P_n^*) = E_n(f)$. Any degree-$(n+1)$ polynomial can be written as $P_{n+1}(x) = a_0 + a_1 x + \cdots + a_{n+1}x^{n+1}$ with coefficient vector $\mathbf{a} \in \mathbb{R}^{n+2}$. Define the error function $\Phi(\mathbf{a}) = \Delta(P_{n+1}) = \sup_{x\in[a,b]} |f(x) - P_{n+1}(x)|$, which is continuous.
Consider the set $K = \mathbf{a} \in \mathbb{R}^{n+2} : \Phi(\mathbf{a}) \leq E_n(f)$. This set is non-empty (since including P_n^* with $a_{n+1} = 0$ gives $\Phi(\mathbf{a}) = E_n(f)$), closed (due to continuity of Φ), and bounded (because bounded error implies bounded coefficients in finite dimensions), hence compact. Since Φ is continuous on K, it attains a minimum at some $\mathbf{a}^* \in K$, s.t. $\Phi(\mathbf{a}^*) = E_{n+1}(f)$. Thus, the polynomial P_{n+1}^* corresponding to $\mathbf{a}^*$ is a degree-$(n+1)$ best approximation polynomial.

Problem 27. (C4S2Q11)
Prove the following statements.

a) A polynomial of odd degree with real coefficients has at least one real root.

b) If P_n is a polynomial of degree n, the function sgn $P_n(x)$ has at most n points of discontinuity.

c) If there are $n+2$ points $x_0 < x_1 < \cdots < x_{n+1}$ in the closed interval $[a,b]$ s.t. the quantity sgn $\left[(f(x_i) - P_n(x_i))(-1)^i\right]$ assumes the same value for $i = 0, \ldots, n+1$, then $E_n(f) \geq \min_{0\leq i\leq n+1} |f(x_i) - P_n(x_i)|$.

(This result is known as Vallee Poussin's theorem.[8] For the definition of $E_n(f)$ see Problem 10.)

Quick Solution. a) $P_{n+1}(x) = a_{n+1}x^{n+1} + Q_n(x)$, as $\lim_{n\to\infty} \frac{Q_n(x)}{x^{n+1}} = 0$, then $\exists$ sufficiently large M, s.t. $P_{n+1}(M) = (a_{n+1} + o(1))M^{n+1} \implies P_{n+1}(M) \cdot P_{n+1}(-M) < 0$, then use Bolzano-Cauchy IVT.

b) Method 1. [Induction]+/differential/+[Contradiction]
Mehtod 2. /Foundamental Theorem of Algebra/
Mehtod 3. /Factorization/

c) (wait to add)

Problem 28. (C4S2Q12)

a) Show that for any $n \in \mathbb{N}$ the function $T_n(x) = \cos(n \arccos x)$ defined on the closed interval $[-1,1]$ is an algebraic polynomial of degree n. (These are the Chebyshev polynomials.)

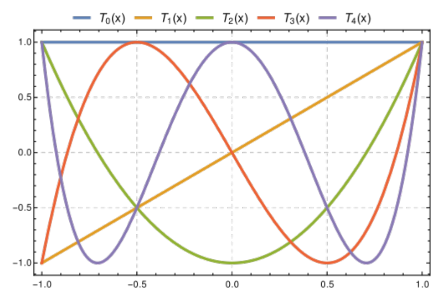
b) Find an explicit algebraic expression for the polynomials T_1, T_2, T_3, and T_4 and draw their graphs.

c) Find the roots of the polynomial $T_n(x)$ on the closed interval $[-1,1]$ and the points of the interval where $|T_n(x)|$ assumes its maximum value.

d) Show that among all polynomials $P_n(x)$ of degree n whose leading coefficient is 1 the polynomial $\frac{1}{2^{n-1}}T_n(x)$ is the unique polynomial closest to zero, that is, $E_n(0) = \frac{1}{2^{n-1}} \max_{|x|\le 1} |T_n(x)| = \frac{1}{2^{n-1}}$. (For the definition of $E_n(f)$ see Problem 10.)

Quick Solution. a) [Induction] $T_1(x) = x,\ T_2(x) = 2x^2-1,\ T_n(x) = 2xT_{n-1}(x) - T_{n-2}$.

b) $T_1(x) = x,\ T_2(x) = 2x^2-1,\ T_3(x) = 4x^3-3x,\ T_4(x) = 8x^4-8x^2+1$.



from Wikipedia

c) Roots: $x_k = \cos\frac{\pi(2k-1)}{2n},\ k \in \mathbb{N}^+$; Max points: $x_k = \cos\frac{k\pi}{n},\ k \in \mathbb{N}$.

d) [Conrtradiction] Let's assume that $w_n(x)$ is a polynomial of degree n with leading coefficient 1 with maximal absolute value on the interval $[-1,1]$ less than $\frac{1}{2^{n-1}}$.
Define $f_n(x) = \frac{1}{2^{n-1}}T_n(x) - w_n(x)$
Because at extreme points of T_n we have $|w_n(x)| < \left|\frac{1}{2^{n-1}}T_n(x)\right|$, and $f_n(x) > 0$ for $x = \cos\frac{2k\pi}{n}$, $f_n(x) < 0$ for $x = \cos\frac{(2k+1)\pi}{n}$.
From the intermediate value theorem, $f_n(x)$ has at least n roots. However, this is impossible, as $f_n(x)$ is a polynomial of degree $n-1$, so FTA implies it has at most $n-1$ roots.

6 Chapter 5.Differential Calculus

6.1 Comprehensions

1st part: Introduce the definition of functions differentiable at a point and related concepts: differential, derivative and increment. Besides, it also includes the geometrical meaning of them and some calculation examples. After basic definitions, we are given the basic rules of differentiation, including: arithmetic operations, differentiation of composite functions, differentiation of inverse functions, and higer order derivatives (Lebniz's formula). And also derivatives of basic elementary functions.
2nd part: Then we dive into some basic theorems of differential calculus: /All of beging-Most basic/ Fermat's Lemma, /Dev on basic-3 Mean Value Theorems/ Rolle's Theorem $\to$ Lagrange's Mean Value Theorem $\to$ Cauchy's Mean Value Theorem (due to last 2 of them associated the increment and derivative, they're also called Theorem of Lagrange and Cauchy on finite increments). Finally, we're given the most important theorem in differential calculus: Taylor's Formula (the reminder term could be obtained using Cauchy's Mean Value Theorem or Theorem of Cauchy on finite increments).
3nd part: It's the last part of this chapter, which focuses on the applications (analyze properties of functions) of differential calculus, including: monotoncity, interior extremum, convacity, and graph sketching. And the well-known L'Hôpital's Rule. Remains lots of examples in Problems on Natural Science.
DLC: The special (which could be called overall) point is that Zorich left a specific section for complex numbers and functions, even including analyticity (or holomorphism).

Definition: /differential/ $\underbrace{\underbrace{f(x+h)-f(x)} = \overbrace{A(x)\underbrace{h}}^{\text{differential}}}_{\text{increments}} + \underbrace{o(a;h)}_{o(h),\ h\to 0}$, $x \in E$

/derivative/ $f'(x) = A(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

All of beging-Most basic

Theorem 6.1. */Fermat's Lemma/*
$f : E \to \mathbb{R}$ is differentiable at an interior extremum $x_0 \in E \implies f'(x_0) = 0$.

3 Mean Value Theorems

Theorem 6.2. */Rolle's theorem/*
$f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b) and $f(a) = f(b)$, then $\exists \xi \in (a,b)$ s.t. $f'(\xi) = 0$.

Theorem 6.3. */Lagrange's Mean Value Theorem/*
$f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then $\exists \xi \in (a,b)$ s.t. $f(b) - f(a) = f'(\xi)(b-a)$.

Theorem 6.4. */Cauchy's Mean Value Theorem/*
Let $x = x(t)$ and $y = y(t)$ be continuous on $[\alpha, \beta]$ and differentiable on (α, β). Then $\exists \tau \in [\alpha, \beta]$ s.t. $x'(\tau)(y(\beta) - y(\alpha)) = y'(\tau)(x(\beta) - x(\alpha))$.
If in addition $x'(t) \neq 0$, $\forall t \in [\alpha, \beta]$, then $x(\alpha) \neq x(\beta)$ and $\frac{y(\beta)-y(\alpha)}{x(\beta)-x(\alpha)} = \frac{y'(\tau)}{x'(\tau)}$.

Master in differential calculus

Theorem 6.5. */Taylor's Formula/*

$$F(a) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + r_n(x_0, x)$$

Remark: /Reminder term/

$$r_n(x_0; x) = \frac{\varphi(x) - \varphi(x_0)}{\varphi'(\xi)n!} f^{(n+1)}(\xi)(x - \xi)^n.$$

or

$$\frac{r_n(x_0; x)}{\varphi(x) - \varphi(x_0)} = \frac{f^{(n+1)}(\xi)(x - \xi)^n}{\varphi'(\xi)n!} \quad \text{/Cauchy's MVT/}$$

/Cauchy's formula of the remainder term/

$$r_n(x_0; x) = \frac{1}{n!} f^{(n+1)}(\xi)(x - \xi)^n (x - x_0).$$

A particularly elegant formula results if we set $\varphi(t) = (x - t)^{n+1}$
/The Lagrange form of the remainder/

$$r_n(x_0; x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x - x_0)^{n+1}.$$

We remark that when $x_0 = 0$ Taylor's formula is often called MacLaurin's formula.

6.2 Problems Sets

Problem 29. (C5S1Q5) /Continuous, but nowhere derivative/

Set $\Psi_0(x) = \begin{cases} x, & \text{if } 0 \le x \le \frac{1}{2}, \\ 1-x, & \text{if } \frac{1}{2} \le x \le 1, \end{cases}$ and extend this function to the entire real line so as to have period 1. We denote the extended function by φ_0. Further, let $\varphi_n(x) = \frac{1}{4^n}\varphi_0(4^n x)$. The function φ_n has period 4^{-n} and a derivative equal to $+1$ or -1 everywhere except at the points $x = \frac{k}{2^{n+1}}, n \in \mathbb{Z}$. Let $f(x) = \sum_{n=1}^{\infty} \varphi_n(x)$.

Show that the function f is defined and continuous on $\mathbb{R}$, but does not have a derivative at any point. (This example is due to the well-known Dutch mathematician B. L. van der Waerden (1903–1996). The first examples of continuous functions having no derivatives were constructed by Bolzano (1830) and Weierstrass (1860).)

Solution. 1. Using Weierstrass M-test (uniform convergence of series of functions) and the uniform limit theorem (the uniform limit of any sequence of continuous functions is continuous), we can show that $f \in C(\mathbb{R})$.
2. Just consider $\lim_{n\to\infty} \frac{f(x\pm 4^{-n})-f(x)}{4^{-n}} = \lim_{n\to\infty} \frac{\sum_{k=0}^{\infty}\varphi_k(x\pm 4^{-n})-\sum_k \varphi_k(x)}{4^{-n}}$, and using the periodicity of φ_k.

/Dev/ **More step problem:**
Prove that the nowhere differential functions are dense in $C(\mathbb{R})$.

Problem 30. (C5S2Q1)

Let $\alpha_0, \alpha_1, \ldots, \alpha_n$ be given real numbers. Find a polynomial $P_n(x)$ of degree n having the derivatives $P_n^{(k)}(x_0) = \alpha_k, k = 0, 1, \ldots, n$, at a given point $x_0 \in \mathbb{R}$.

Quick Solution. $P_n(x) = \sum_{k=0}^{n} \alpha_k (x - x_0)^k$

Problem 31. (C5S2Q2)

Quick Solution. a) $f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}, f'(x) = \begin{cases} \frac{2e^{-\frac{1}{x^2}}}{x^3} & x \neq 0 \\ 0 & x = 0 \end{cases},$

$f^{(n)}(x) = \lim_{x\to\infty} \frac{f^{(n-1)}(x)-0}{x-0} = [\text{Induction}] = \lim_{x\to\infty} \frac{R_n(x)}{e^{\frac{1}{x^2}}} = 0.$

b) $f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}, f(x) = \begin{cases} 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$

c) $f(x) = \begin{cases} e^{-\frac{1}{(1+x)^2}} \cdot e^{-\frac{1}{(1-x)^2}} & -1 < x < 1 \\ 0 & 1 \le |x| \end{cases}$

Problem 32. (C5S2Q3)

$$\left[x^{n-1} f\left(\frac{1}{x}\right)\right]^{(n)} = \frac{(-1)^n}{x^{n+1}} f^{(n)}\left(\frac{1}{x}\right)$$

Quick Solution. [Induction]. Assume that the base case $n = 1$ is proved. Assuming that $\frac{d^n}{dx^n}\left[x^{n-1}f\left(\frac{1}{x}\right)\right] = \frac{(-1)^n}{x^{n+1}}f^{(n)}\left(\frac{1}{x}\right)$, we have

$$\frac{d^{n+1}}{dx^{n+1}}\left[x^n f\left(\frac{1}{x}\right)\right] = \frac{d^n}{dx^n}\left[nx^{n-1}f\left(\frac{1}{x}\right) - x^{n-2}f'\left(\frac{1}{x}\right)\right]$$

$$= \underbrace{\frac{n(-1)^n}{n^{n+1}}f^{(n)}\left(\frac{1}{x}\right)}_{\text{by asumption}} - \frac{d}{dx}\left(\frac{d^{n-1}}{dx^{n-1}}\left[x^{n-2}f'\left(\frac{1}{x}\right)\right]\right)$$

$$= \frac{n(-1)^n}{n^{n+1}}f^{(n)}\left(\frac{1}{x}\right) - \frac{d}{dx}\underbrace{\left[\frac{(-1)^{n-1}}{x^n}f^{(n)}\left(\frac{1}{x}\right)\right]}_{\text{by assumption}}$$

$$= \frac{n(-1)^n}{n^{n+1}}f^{(n)}\left(\frac{1}{x}\right) + \frac{n(-1)^{n-1}}{x^{n+1}}f^{(n)}\left(\frac{1}{x}\right) + \frac{(-1)^{n+1}}{x^{n+2}}f^{(n+1)}\left(\frac{1}{x}\right)$$

$$= \frac{(-1)^{n+1}}{x^{n+2}}f^{(n+1)}\left(\frac{1}{x}\right) \square$$

And that about wraps it up. *"from AOPS"*

Problem 33. (C5S2Q4)
Let f be a differentiable function on $\mathbb{R}$. Show that

a) if f is an even function, then f' is an odd function;

b) if f is an odd function, then f' is an even function;

b) (f' is odd) $\Leftrightarrow$ (f is even).

Quick Solution. By definition.

Problem 34. (C5S2Q5)
Show that

a) the function $f(x)$ is differentiable at the point x_0 if and only if $f(x) - f(x_0) = \varphi(x)(x - x_0)$, where $\varphi(x)$ is a function that is continuous at x_0 (and in that case $\varphi(x_0) = f'(x_0)$);

b) if $f(x) - f(x_0) = \varphi(x)(x - x_0)$ and $\varphi \in C^{(n-1)}(U(x_0))$, where $U(x_0)$ is a neighborhood of x_0, then $f(x)$ has a derivative $(f^{(n)}(x_0))$ of order n

Quick Solution. By definition.

Problem 35. (C5S2Q6)
Give an example showing that the assumption that f^{-1} be continuous at y_0 cannot be omitted from following theorem:

Theorem 6.6. *(The derivative of an inverse function). Let the functions $f: X \to Y$ and $f^{-1}: Y \to X$ be mutually inverse and continuous at points $x_0 \in X$ and $f(x_0) = y_0 \in Y$ respectively. If f is differentiable at x_0 and $f'(x_0) \neq 0$, then f^{-1} is also differentiable at the point y_0, and $(f^{-1})'(y_0) = (f'(x_0))^{-1}$.*

Quick Solution. Let $f(x) = \begin{cases} x + x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, after calculating, we could find f is continuous and differentiable, but f^{-1} doesn't even exist.

Problem 36. (C5S3Q1)
Choose numbers a and b so that the function $f(x) = \cos x - \frac{1+ax^2}{1+bx^2}$ is an infinitesimal of highest possible order as $x \to 0$.

Quick Solution. $b = 0,\ a = -\frac{1}{2}$ $(\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 + \mathcal{O}(x^6))$

Problem 37. (C5S3Q2)

$$\lim_{x\to\infty} x\left[\frac{1}{e} - (\frac{x}{x+1})^x\right]$$

Solution.

$$\lim_{x\to\infty} x\left[\frac{1}{e} - (\frac{x}{x+1})^x\right] = \lim_{x\to\infty} x\left[\frac{1}{e} - e^{x\ln(\frac{x}{x+1})}\right] \quad (\ln\frac{x}{x+1} = \ln(1 - \frac{1}{x+1}))$$

$$= \lim_{x\to\infty} x\left[e^{-1} - e^{x(-\frac{1}{x+1} - \frac{1}{2(x+1)^2} + \mathcal{O}(\frac{1}{x^3}))}\right]$$

$$= \lim_{x\to\infty} x\left[e^{-1} - e^{-\frac{x}{x+1} - \frac{x}{2(x+1)^2} + \mathcal{O}(\frac{1}{x^2})}\right]$$

$$= \lim_{x\to\infty} x\left[e^{-1} - e^{-\frac{x}{x+1}}(1 - \frac{x}{2(x+1)^2} + \mathcal{O}(\frac{1}{x^2}))\right]$$

$$= \lim_{x\to\infty} e^{-1}\left[x - xe^{\frac{1}{x+1}} + \frac{x^2}{2(x+1)^2}e^{\frac{1}{x+1}} + \mathcal{O}(\frac{1}{x})\right]$$

$$= \lim_{x\to\infty} e^{-1}\left[x(1 - (1 + \frac{1}{x+1} + \mathcal{O}(\frac{1}{x^2}))) + \frac{x^2}{2(x+1)^2}e^{\frac{1}{x+1}} + \mathcal{O}(\frac{1}{x})\right]$$

$$= \lim_{x\to\infty} e^{-1}\left[\frac{x}{x+1} + \frac{x^2}{2(x+1)^2}(1 + o(1)) + \mathcal{O}(\frac{1}{x})\right]$$

$$= \frac{3}{2}e^{-1}$$

Problem 38. (C5S3Q3)

Quick Solution. $e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots$

Problem 39. (C5S3Q4)
Let f be a function that is infinitelt differentiable at 0. Show that:

a) if f is even, then its Taylor series at 0 contains only even powers of x;

b) if f is odd, then its Taylor series at 0 contains only odd powers of x.

Quick Solution. The parity of f and f' is opposite. & When $f(x) = -f(-x)$, $f(0) = 0$.

Problem 40. (C5S3Q5)
Show that if $f \in C^{(\infty)}[-1,1]$ and $f^{(n)}(0) = 0$ for $n = 0, 1, 2, \ldots$, and there $\exists$ a number C s.t. $\sup_{x\in[-1,1]} |f^{(n)}(x)| \le n!C$ for $n \in \mathbb{N}$, then $f = 0$ on $[-1,1]$.

Quick Solution. Taylor's formula: on $x = 0$, $|f(x)| = |\frac{f^{(n+1)}(\varepsilon_n)}{(n+1)!}x^{n+1}| \le |\frac{(n+1)!C}{(n+1)!}x^{n+1}| = |Cx^{n+1}| = \mathcal{O}(x^n)$. Since $f(0) = 0$, and continuity, $f(x) \equiv 0$.

Problem 41. (C5S3Q6)
Let $f \in C^{(n)}(-1,1)$ and $\sup\limits_{-1<x<1} |f(x)| \le 1$. Let $m_k(I) = \inf\limits_{x\in I} |f^{(k)}(x)|$, where I is an interval contained in $(-1,1)$. Show that

a) if I is partitioned into three successive intervals I_1, I_2, and I_3 and μ is the length of I_2, then $m_k(I) \le \frac{1}{\mu}\left(m_{k-1}(I_1) + m_{k-1}(I_3)\right)$;

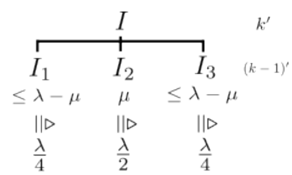
b) if I has length λ, then $m_k(I) \le \frac{2^{k(k+1)/2}k^k}{\lambda^k}$;

c) there exists a number α_n depending only on n such that if $|f'(0)| \ge \alpha_n$, then the equation $f^{(n)}(x) = 0$ has at least $n-1$ distinct roots in $(-1,1)$.

Hint: In part b) use part a) and mathematical induction; in c) use a) and prove by induction that there exists a sequence $x_{k_1} < x_{k_2} < \cdots < x_{k_k}$ of points of the open interval $(-1,1)$ s.t. $f^{(k)}(x_{k_i}) \cdot f^{(k)}(x_{k_{i+1}}) < 0$ for $1 \le i \le k-1$.

Solution. a) (Lagrange MVT)
$m_k(I) = \inf\limits_{x\in I\subset(-1,1)} |f^{(k)}(x)| \le m_k(I_2) = \inf\limits_{x\in I_2\subset I} |f^{(k)}(x)| \le \forall |f^k(x)| \in \{|f^k(x)| : x \in I_2\} = [|f^k(\xi)|, \xi \in I_2] = \frac{\inf\limits_{x\in I_1\subset I} |f^{(k-1)}(x)| - \inf\limits_{x\in I_3\subset I} |f^{(k-1)}(x)|}{x_1 - x_2} \le \frac{m_{k-1}(I_1)+m_{k-1}(I_3)}{\mu}$

b) $m_k(I) = \inf\limits_{x\in I} |f^{(k)}(x)|$, $\mu(I) = \lambda \le 2$
$m_k(I) \le \frac{1}{\mu}(m_{k-1}(I_1) + m_{k-1}(I_3)) = [\mu = \mu(I_2) \triangleq \frac{\lambda}{2}, \mu(I_1) = \mu(I_3) = \frac{\lambda}{4}] = \frac{2}{\lambda}(m_{k-1}(I_1) + m_{k-1}(I_3)) \le \frac{2\cdot 1}{\lambda} \cdot \frac{4\cdot 2}{\lambda} \max\{m_{k-2}(I_{1,1}) + m_{k-2}(I_{1,3}), m_{k-2}(I_{3,1}) + m_{k-2}(I_{3,3})\} = \cdots = \frac{2^1\cdot 1}{\lambda} \cdot \frac{2^2\cdot 2}{\lambda} \cdots \frac{2^k\cdot 2}{\lambda} \cdot 2 \sup\limits_{-1<x<1} |f(x)| \le \frac{2^{\frac{k(1+k)}{2}}}{\lambda^k} \cdot 2^k \le \frac{2^{\frac{k(1+k)}{2}}}{\lambda^k} \cdot k^k$, $(k \le 2)$



[Cantor's Set-Method]

c) (Wait to prove)

Problem 42. (C5S3Q7)
Show that if a function f is defined and differentiable on an open interval I and $[a,b] \subset I$, then

a) the function $f'(x)$ (even if it is not continuous!) assumes on $[a,b]$ all the values between $f'(a)$ and $f'(b)$ (the theorem of Darboux);

b) if $f''(x)$ also exists in (a,b), then there is a point $\xi \in (a,b)$ such that $f'(b) - f'(a) = f''(\xi)(b-a)$.

Quick Solution. a) (Fermat's Lemma + Weierstrass Maximum Theorem) /Constructive Proof/
Let y be an arbitrary number between $f'(a)$ and $f'(b)$, and consider the function $F(x) = f(x) - yx$. Then $F'(x) = f'(x) - y$, and $F'(a) = f'(a) - y$, $F'(b) = f'(b) - y$. $F'(a) \cdot F'(b) < 0$.

b) (Lagrange's MVT)

Lemma. f' doesn't have the 1st kind of discontinuity.

The proof for Lemma is very easy, just choose a small neibourhood of discontinuity and using a).
Apply Lagrange's MVT to f' on $[a,b]$, we could get the result.

Problem 43. (C5S3Q8)
A function $f(x)$ may be differentiable on the entire real line, without having a continuous derivative $f'(x)$

a) Show that $f'(x)$ can have only discontinuities of second kind.

b) Find the flaw in the following "proof" that $f'(x)$ is continuous:
Proof. Let x_0 be an arbitrary point on $\mathbb{R}$ and $f'(x_0)$ the derivative of f at the point x_0. By definition of the derivative and Lagrange's theorem $f'(x_0) = \lim_{x \to x_0} \frac{f(x)-f(x_0)}{x-x_0} = \lim_{x \to x_0} f'(\xi) = \lim_{\xi \to x_0} f'(\xi)$, where ξ is a point between x_0 and x and therefore tends to x_0 as $x \to x_0$.

Quick Solution. a) Proof is aformentioned.

b) Lagrange's MVT requires the contimuity of function on closed interval.

Problem 44. (C5S3Q9)
Let f be twice differentiable on an interval I. Let $M_0 = \sup_{x\in I}|f(x)|$, $M_1 = \sup_{x\in I}|f'(x)|$ and $M_2 = \sup_{x\in I}|f''(x)|$. Show that

a) if $I = [-a,a]$, then $|f'(x)| \le \frac{M_0}{a} + \frac{x^2+a^2}{2a}M_2$;

b) $\begin{cases} M_1 \le 2\sqrt{M_0M_2}, & \text{if the length of } I \text{ is not less than } 2\sqrt{M_0/M_2}, \\ M_1 \le \sqrt{2M_0M_2}, & \text{if } I = \mathbb{R}; \end{cases}$

c) the numbers 2 and $\sqrt{2}$ in part b) cannot be replaced by smaller numbers;

d) if f is differentiable p times on $\mathbb{R}$ and the quantities M_0 and $M_p = \sup_{x\in\mathbb{R}}|f^{(p)}(x)|$ are finite, then the quantities $M_k = \sup_{x\in\mathbb{R}}|f^{(k)}(x)|$, $1 \le k < p$, are also finite and $M_k \le 2^{k(p-k)/2}M_0^{1-k/p}M_p^{k/p}$.

Hint: Use Exercises 9b) and mathematical induction.

Solution. a) Using Taylor's expansion: $f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2}(x-x_0)^2$ at $x = a$ and $x = -a$:
$f(a) = f(x_0) + f'(x_0)(a-x_0) + \frac{f''(\xi_1)}{2}(a-x_0)^2$, $f(-a) = f(x_0) + f'(-a)(-a-x_0) + \frac{f''(\xi_2)}{2}(-a-x_0)^2$.
Then $f'(x_0) = \frac{f(a)-f(-a)}{2a} - \frac{1}{4a}f''(\xi_1)(a-x_0)^2 + \frac{1}{4a}f''(\xi_2)(a+x_0)^2$, using triangle inequality gives: $|f'(x)| \le \frac{M_0}{a} + \frac{M_2}{4a}(a-x_0)^2 + \frac{M_2}{4a}(a+x_0)^2 = \frac{M_0}{a} + \frac{M_2}{2a}(a^2+x_0^2)$.

b) i) Consider $I = [a,b]$, the method is absolutely same as a).
ii) Perform the analysis again for some intervals $[x-a,x)$, $(x,x+a]$ and we recover a similar result as above: $f(x-a) = f(x) + f'(x)(-a) + \frac{f''(c_1)}{2}(a)^2$, $f(x+a) = f(x) + f'(x)(a) + \frac{f''(c_2)}{2}(a)^2$.
Subtracting the two Taylor theorem estimates we obtain that: $f'(x) = \frac{f(x+a)-f(x-a)}{2a} - \frac{f''(c_1)+f''(c_2)}{4a}a^2$ and we have that by triangle inequality: $|f'(x)| \le \frac{M_0}{a} + \frac{M_2 a}{2}$ Since $|f'(x)| \le \delta(a) \quad \forall a \in \mathbb{R}$ it must be true that since $|f'(x)| \le M_1$ for all a as well, we must have that $M_1 \le \inf_a \delta(a)$ for if the opposite happens, then there will be some subset of the reals where the function violates the first inequality. We compute the infimum as follows: $\delta(a) = \frac{M_0}{a} + \frac{a}{2}M_2 \ge \frac{M_0}{a} + \frac{a}{2}M_2 \ge 2\sqrt{\frac{M_0M_2}{2}} = \sqrt{2M_0M_2}$ and the proposition is proven. The infimum happens when $a = \sqrt{\frac{2M_0}{M_2}}$.

c) Example: $f(x) = 2x^2 - 1 - a^2, \quad I = [a,1]$, $\forall a \in [0,1)$.

d) [Induction]
$p = 2$, $M_1 \le 2^{\frac{1}{2}}M_0^{\frac{1}{2}}M_2^{\frac{1}{2}} = \sqrt{2M_{0M_2}}$ ✓;
$p = 2$, suppose p holds $\forall 1 \le k \le p-1$: $\underline{M_k \le 2^{\frac{k(p-K)}{2}}M_0^{1-\frac{k}{p}}M_p^{\frac{k}{p}}}$. Then for $p+1$: first, using MVT(Taylor's expansion is allright)

$$\begin{cases} f^{(p-1)}(x+h) = f^{(p-1)}(x) + hf^p(x) + \frac{h^2}{2}f^{(p+1)}(\xi_1) \\ f^{(p-1)}(x+h) = f^{(p-1)}(x) + hf^p(x) + \frac{h^2}{2}f^{(p+1)}(\xi_1) \end{cases},$$
then $f^p(x) = \frac{1}{2h}(f^{(p-1)}(x+h) - f^{(p-1)}(x-h)) - \frac{h}{4}[f^{(p+1)}(\xi_1) - f^{(p+1)}(\xi_2)]$, then $M_p \leq \frac{1}{h}M_{p-1} + \frac{h}{2}M_{p+1} = \left[h = \sqrt{2\frac{M_{p-1}}{M_{p+1}}}\right] = \sqrt{2M_{p-1}M_{p+1}}$, i.e. $\underline{M_p \leq 2^{\frac{1}{2}}M_{p-1}^{\frac{1}{2}}M_{p+1}^{\frac{1}{2}}}$.

Using induction hypothesis, we have $\underline{M_{p-1} \leq 2^{\frac{p-1}{2}}M_0^{\frac{1}{p}}M_p^{\frac{1-p}{p}}}$, conbining 2 of them, then $\underline{M_p \leq 2^{\frac{p}{2}}M_0^{\frac{1}{p+1}}M_{p+1}^{\frac{p}{p+1}}}$, then we could verify the condition for $p+1$.

Problem 45. (C5S3Q10)
Show that if a function f has derivatives up to order $n+1$ inclusive at a point x_0 and $f^{(n+1)}(x_0) \neq 0$, then in the Lagrange form of the remainder in Taylor's formula $r_n(x_0; x) = \frac{1}{n!}f^{(n)}(x_0 + \theta(x-x_0))(x-x_0)^n$, where $0 < \theta < 1$ and the quantity $\theta = \theta(x)$ tends to $\frac{1}{n+1}$ as $x \to x_0$.

Quick Solution. Just compare $f(x) = \sum_{k=0}^{n-1}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \frac{f^{(n)}(x_0+\theta_x(x-x_0))}{n!}(x-x_0)^n$ and $f(x) = \sum_{k=0}^{n-1}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(x_0+\phi_x(x-x_0))}{(n+1)!}(x-x_0)^{n+1}$, then we get $\frac{f^{(n)}(x_0+\theta_x(x-x_0))}{n!} = \frac{f^{(n)}(x_0)}{n!} + \frac{f^{(n+1)}(x_0+\phi_x(x-x_0))}{(n+1)!}(x-x_0)$ i.e. $\frac{f^{(n)}(x_0+\theta_x(x-x_0)) - f^{(n)}(x_0)}{x-x_0} = \frac{f^{(n+1)}(x_0+\phi_x(x-x_0))}{n+1}$, then $\theta_x = \frac{1}{n+1}\cdot\frac{f^{(n+1)}(x_0+\phi_x(x-x_0))}{\underline{\frac{f^{(n)}(x_0+h_x) - f^{(n)}(x_0)}{h_x}}}$, where $h_x = \theta_x(x-x_0)$. Thus $\lim_{X\to x_0}\theta_x = \frac{1}{n+1}$.

Problem 46. (C5S4Q1)
Let $x = (x_1, \ldots, x_n)$ and $\alpha = (\alpha_1, \ldots, \alpha_n)$, where $x_i \geq 0$, $\alpha_i > 0$ for $i = 1, \ldots, n$ and $\sum_{i=1}^n \alpha_i = 1$. For any number $t \neq 0$ we consider the mean of order t of the numbers $x_1, \ldots, x_n$ with weights α_i: $M_t(x,\alpha) = \left(\sum_{i=1}^n \alpha_i x_i^t\right)^{1/t}$. In particular, when $\alpha_1 = \cdots = \alpha_n = \frac{1}{n}$, we obtain the harmonic, arithmetic, and quadratic means for $t = -1, 1, 2$ respectively. Show that

a) $\lim_{t\to 0} M_t(x,\alpha) = x_1^{\alpha_1}\cdots x_n^{\alpha_n}$, that is, in the limit one can obtain the geometric mean;

b) $\lim_{t\to+\infty} M_t(x,\alpha) = \max_{1\leq i\leq n} x_i$;

c) $\lim_{t\to-\infty} M_t(x,\alpha) = \min_{1\leq i\leq n} x_i$;

d) $M_t(x,\alpha)$ is a nondecreasing function of t on $\mathbb{R}$ and is strictly increasing if $n > 1$ and the numbers x_i are all nonzero.

Solution. a) Attempt 1. (Convacity of $\ln x$) $\lim_{t\to 0}(\sum\limits_{i=1}^{n}\alpha_i x_i^t)^{\frac{1}{t}} = \lim_{t\to 0} e^{\frac{1}{t}\ln(\sum\limits_{i=1}^{n}\alpha_i x_i^t)} \geq \lim_{t\to 0} e^{\frac{1}{t}\sum\limits_{i=1}^{n}\alpha_i \ln x_i^n} = \prod_{i=1}^{n} x_i^{\alpha_i}$

Attempt 2. (L'Hôpital) $\lim_{t\to 0}(\sum\limits_{i=1}^{n}\alpha_i x_i^t)^{\frac{1}{t}} = \lim_{t\to 0} e^{\frac{1}{t}\ln(\sum\limits_{i=1}^{n}\alpha_i x_i^t)} = e^{\lim\limits_{t\to 0}\frac{\ln(\sum\limits_{i=1}^{n}\alpha_i x_i^t)}{t}} = e^{\lim\limits_{t\to 0}\frac{\sum\limits_{i=1}^{n}\alpha_i x_i^t \ln x_i}{\sum\limits_{i=1}^{n}\alpha_i x_i^t}} = e^{\sum\limits_{i=1}^{n}\alpha_i \ln x_i} = \prod_{i=1}^{n} x_i^{\alpha_i}$

b) Let $\max_{1\leq i\leq n}\{x_i\} = x_k$, then $\lim_{t\to\infty}(\alpha_k x_k^t)^{\frac{1}{t}} \leq \lim_{t\to\infty} M_t(x,\alpha) \leq \lim_{t\to\infty}((\sum\limits_{i=1}^{n}\alpha_i)x_k^t)^{\frac{1}{t}}$.

c) Let $\min_{1\leq i\leq n}\{x_i\} = x_l$, then $\lim_{t\to\infty}(\alpha_k x_l^t)^{\frac{1}{t}} \geq \lim_{t\to\infty} M_t(x,\alpha) \geq \lim_{t\to\infty}((\sum\limits_{i=1}^{n}\alpha_i)x_l^t)^{\frac{1}{t}}$.

d) /Jensen's Inequality/ Notice that $y = x^m,\ x\geq 0$ is convex,then $(\sum_{i=1}^{n}\alpha_i x_i^s)^{\frac{t}{s}} \leq \sum_{i=1}^{n}\alpha_i (x_i^s)^{\frac{t}{s}} = \sum_{i=1}^{n}\alpha_i x_i^t \implies (\sum_{i=1}^{n}\alpha_i x_i^s)^{\frac{1}{s}} \leq (\sum_{i=1}^{n}\alpha_i x_i^t)^{\frac{1}{t}}$. Then $\forall s\leq t\in\mathbb{R}$, $M_s(x,\alpha)\leq M_t(x,\alpha)$. "=" i.f.f. $\forall x_i,\ i\in\{1,2,\cdots,n\}$, x_i all equals (It's waiting to prove).

Problem 47. (C5S4Q2)
Show that $|1+x|^p \geq 1+px+c_p\varphi_p(x)$, where c_p is a constant depending only on p, $\varphi_p(x) = \begin{cases} |x|^2 \text{ for } |x|\leq 1, \\ |x|^p \text{ for } |x|>1, \end{cases}$, if $1<p\leq 2$, and $\varphi_p(x) = |x|^p$ on $\mathbb{R}$ if $2<p$.

Solution. $\boxed{\text{Lemma. } (1+x)^p \geq 1+px \quad \forall x\neq 0,\ p>1.}$

1°. $1<p\leq 2$, $|x|\leq 1$. /Cauchy's MVT/

$$\begin{aligned}\frac{(1+x)^{p-(1+px)}}{x^2} &= \frac{(p(1+x_1)^{p-1}-p)}{2x_1} \quad \exists x_i\in(-1,1)/\{0\} \\ &= \frac{p(p-1)(1+x_2)^{p-2}}{2} \quad \exists x_2\in(-x_1,x_1)/\{0\} \\ &\geq \frac{p(p-1)}{2}\end{aligned}$$

then $(1+x)^p \geq 1+px+\frac{p(p-1)}{2}x^2$.

2°. $1<p\leq 2$, $|x|>1$.

Let $g(x) = \frac{(1+x)^p-1-px}{|x|^p}$ is continuous and $g(x)\leq 0$ on $(-\infty,-1]\cup[1,\infty)$. $g(1) = \lim_{x\to 1^+} g(x) = \frac{2^p-(1+p)}{1} > 0$, $g(-1) = \lim_{x\to 1^-} g(x) = \frac{p-1}{1} > 0$, $\lim_{x\to+\infty} g(x) = \lim_{x\to-\infty} g(x) = 1$. Then $g(x)>0$ on $(-\infty,-1)\cup(1,\infty)$, i.e. $\exists c_p>0$ s.t. $g(x) = \frac{(1+x)^p-1-px}{|x|^p} \geq c_p$, i.e. $(1+x)^p \geq 1+px+c_p|x|^p$.

3°. $p > 2,\ x \in \mathbb{R}$
Analysis is similiar with 2°. Let $g(x) = \frac{(1+x)^p - 1 - px}{|x|^p}$ and analyze its boundary. (Maybe L'Hospital Rule is used in process.)

Problem 48. (C5S4Q3)
Verify that $\cos x < (\frac{\sin x}{x})^3$, for $0 < |x| < \frac{\pi}{2}$.

Quick Solution. Let $f(x) = \sin x(\cos x)^{-\frac{1}{3}} - x$, $x \in (0, \frac{\pi}{2})$ ($x \in (-\frac{\pi}{2}, 0)$ is same).

$$\begin{aligned} f'(x) &= \cos^{\frac{2}{3}} x + \frac{1}{3}\sin^2 x \cos^{-\frac{4}{3}} x - 1 \\ &= \cos^{\frac{2}{3}} x + \frac{1}{3}\cos^{-\frac{4}{3}} x - \frac{1}{3}\cos^{\frac{2}{3}} x - 1 \\ &= (\frac{1}{3}\cos^{-\frac{4}{3}} x - \frac{2}{3}\cos^{-\frac{2}{3}} x + \frac{1}{3}) + \frac{2}{3}(\cos^{\frac{2}{3}} x - 2 + \cos^{-\frac{2}{3}} x) \\ &= \frac{1}{3}(\cos^{-\frac{2}{3}} x - 1)^2 + \frac{1}{3}(\cos^{\frac{1}{3}} x - \cos^{-\frac{1}{3}} x)^2 \geq 0 \end{aligned}$$

Problem 49. (C5S4Q5)
Show that if $f \in C \cap (a, b)$ and the inequality $f\left(\frac{x_1+x_2}{2}\right) \leq \frac{f(x_1)+f(x_2)}{2}$ holds for any points $x_1, x_2 \in (a, b)$, then the function f is convex on (a, b).

Solution. Main method is [(Cauchy's Forward or Backward) Induction], first prove 2 lemma:

Lemma 1. $f(\frac{x_1+x_2+\cdots+x_{2^n}}{2^n}) \leq \frac{f(x_1)+f(x_2)+\cdots+f(x_{2^n})}{2^n}$.

[Induction]

Lemma 2. If $f(\frac{x_1+\cdots+x_{n+1}}{n+1}) \leq \frac{f(x_1)+\cdots+f(x_{n+1})}{n+1}$ holds, then $f(\frac{x_1+\cdots+x_n}{n}) \leq \frac{f(x_1)+\cdots+f(x_n)}{n}$.

Then $\forall k \in \mathbb{Z}^+$, $f(\frac{x_1+\cdots+x_k}{k}) \leq \frac{f(x_1)+\cdots+f(x_k)}{k}$, $\forall x_i \in (a, b)$.
Let $k = q$, $\forall 0 < p < q \in \mathbb{Z}^+$, let $x_1 = \cdots = x_p = x'_p,\ x_{p+1} = \cdots = x_k = x'_2$, then $f(\frac{p}{q}x'_1 + (1-\frac{p}{q})x'_2) \leq \frac{p}{q}f(x'_1) + (1-\frac{p}{q})f(x'_2)$.
Since $f \in C(a, b)$, then $\forall \lambda \in (0, 1)$, $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$, i.e. f is convex on (a, b).

Problem 50. (C5S4Q6)
Shaw that

a) if a convex function $f : \mathbb{R} \to \mathbb{R}$ is bounded, it is constant;

b) if $\lim\limits_{x\to-\infty} \frac{f(x)}{x} = \lim\limits_{x\to+\infty} \frac{f(x)}{x} = 0$, for a convex function $f : \mathbb{R} \to \mathbb{R}$, then f is constant.

c) for any convex function f defined on an open interval $a < x < +\infty$ (or $-\infty < x < a$), the ratio $\frac{f(x)}{x}$ tends to a finite limit or to infinity as x tends to infinity in the domain of definition of the function.

Solution. b) $\boxed{\text{Lemma. If } f : D \to \mathbb{R} \text{ is convex, then } \forall x_0 \le x \le y \in D\text{, satisfy } \frac{f(x)-f(x_0)}{x-x_0} \le \frac{f(y)-f(x_0)}{y-x_0}.}$

Proof for lemma: let $f(x) = f(\alpha_1 x_0 + \alpha_2 y) \le \alpha_1 f(x_0) + \alpha_2 f(y)$ and $x - x_0 = \alpha_2(y - x_0)$.
Using Lemma, let $x_0 = 0$, then $\frac{f(y)-f(0)}{y} \le \frac{f(x)-f(0)}{x}$, $y \le x,\ x,\ y \ne 0$
$\lim_{y\to-\infty} \frac{f(y)-f(0)}{y} = 0 \ge \frac{f(x)-f(0)}{x}$, $\forall 0 < x \le y$, then $f(x) \le f(0)$, $x > 0$,
$\lim_{x\to-\infty} \frac{f(x)-f(0)}{x} = 0 \le \frac{f(y)-f(0)}{y}$, $\forall x < 0 < y$, then $f(y) \le f(0)$, $y > 0$,
Similiar with $x < 0$, then $f(x) \equiv f(0)$, $\forall x \in \mathbb{R}$.

a) Method 1. By b), it's easy to prove.
Method 2. [Contradiction] (Withoud b))
If not, $\exists x_1 < x_2 \in \mathbb{R}$ s.t. $f(x_1) < f(x_2)$ ($f(x_1) > f(x_2)$ is same), then $\forall \alpha_1 + \alpha_2 = 1, \exists x_3 > x_2$ s.t. $x_2 = \alpha_1 x_1 + \alpha_2 x_3$, then $f(x_2) \le \alpha_1 f(x_1) + \alpha_2 f(x_3)$, i.e. $\Pi: \begin{cases} \frac{\alpha_1}{\alpha_2}[f(x_2) - f(x_1)] \le f(x_3) - f(x_2) \\ \cdots \text{ go on} \\ \frac{\alpha_1}{\alpha_2} f(x_n) - f(x_{n-1}) \le f(x_{n+1}) - f(x_n) \end{cases}$. Let $\frac{\alpha_1}{\alpha_2} > 1$, then $\left(\frac{\alpha_1}{\alpha_2}\right)^n [f(x_2) - f(x_1)] \le f(x_{n+1}) - f(x_n)$, i.e. $\exists \{x_n\}_{n=1}^{+\infty} \nearrow$, $f(x_{n+1}) \to +\infty$ as $n \to +\infty$, contradicting the boundedness of f.

c) Still using b). lemma, construct $g(x) \triangleq \frac{f(x)-f(a)}{x-a}$, $x \in (a, +\infty)$ (define $f(a) = \lim_{x\to a^+} f(x)$). Then $g(x) \nearrow$, $x > a$, with Weierstrass Theorem, $\lim_{x\to+\infty} g(x) = c\ (\le \infty)$. For $f(x)$, $x \le -a$, $g(x)$ is similar.

Problem 51. (C5S4Q7)
Show that if $f : (a, b) \to \mathbb{R}$ is a convex function, then

a) at any point $x \in (a, b)$ it has a left-hand derivative f'_- and a right-hand derivative f'_+, defined as $f'_-(x) = \lim\limits_{h\to-0} \frac{f(x+h)-f(x)}{h}$, $f'_+(x) = \lim\limits_{h\to+0} \frac{f(x+h)-f(x)}{h}$, and $f'_-(x) \le f'_+(x)$;

b) the inequality $f'_+(x_1) \le f'_-(x_2)$ holds for $x_1, x_2 \in (a, b)$ and $x_1 < x_2$;

c) the set of cusps of the graph of $f(x)$ (for which $f' - (x) \ne f' + (x)$) is at most countable.

Solution. a) Method 1. [Proof by contradiction] Using definition, it's easy to get contradiction.
Mathod 2. Using lemma in (C5S4Q6)b), direct result obtained.

b) $\boxed{\text{Lemma. (Developement in (C5S4Q6)b)) If } f : D \to \mathbb{R} \text{ is convex, then } \forall x \in D,\ \forall x, y : x \le x_0 \le y \in D \text{, s.t. } \frac{f(x)-f(x_0)}{x-x_0} \left(\le \frac{f(y)-f(x_0)}{y-x_0}\right) \le \frac{f(y)-f(x)}{y-x}.}$

b) Proof for Lemma: similiar with (C5S4Q6)b). (using twice).
$\forall x_1 < x_2 \in (a,b)$, $x_1 < x_2$, $\exists h_+ > 0, h_- < 0$, s.t. $x_1 < x_1 + h_+ < x_2 + h_- < x_2$. Then using twice of Lemma, we could have $\frac{f(x_1+h_+)-f(x_1)}{h_+} \le \frac{f(x_2+h_-)-f(x_1+h_+)}{x_2+h_- -(x_1+h_+)} \le \frac{f(x_2)-f(x_2+h_-)}{-h_-} \implies f'_+(x_1) \le f'_-(x_2)$.

c) The proof is similiar with the one of "The first kind of discontinuities of a monotone function are at most countable".
Proof idea and recall: $\forall x_0 \in \{x : f'_-(x) \neq f'_+(x)\} = \{x : f'_-(x) < f'_+(x)\}$, $\exists$ an open interval $(f'_-(x_0), f'_+(x_0))$ corresponding to x_0. And due to "The number of open intervals on the real number axis that do not intersect each other is at most countable", then problem is solved.

Problem 52. (C5S4Q8)
The Legendre transform of a function $f : I \to \mathbb{R}$ defined on an interval $I \subseteq \mathbb{R}$ is the function $f^*(t) = \sup\limits_{x \in I}(tx - f(x))$.
Show that

a) The set I^* of values of $t \in \mathbb{R}$ for which $f^*(t) \in \mathbb{R}$ (that is, $f^*(t) \neq \infty$) is either empty or consists of a single point, or is an interval of the line, and in this last case the function $f^*(t)$ is convex on I^*.

b) If f is a convex function, then $I^* \neq \emptyset$, and for $f^* \in C(I^*)$ $(f^*)^* = \sup\limits_{t \in I^*}(xt - f^*(t)) = f(x)$ for any $x \in I$. Thus the Legendre transform of a convex function is involutive, (its square is the identity transform).

c) The following inequality holds: $xt \le f(x) + f^*(t)$ for $x \in I$ and $t \in I^*$.

d) When f is a convex differentiable function, $f^*(t) = tx_t - f(x_t)$, where x_t is determined from the equation $t = f'(x)$. Use this relation to obtain a geometric interpretation of the Legendre transform f^* and its argument t, showing that the Legendre transform is a function defined on the set of tangents to the graph of f.

e) The Legendre transform of the function $f(x) = \frac{1}{\alpha}x^\alpha$ for $\alpha > 1$ and $x \ge 0$ is the function $f^*(t) = \frac{1}{\beta}t^\beta$, where $t \ge 0$ and $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. Taking account of c), use this fact to obtain Young's inequality, which we already know: $xt \le \frac{1}{\alpha}x^\alpha + \frac{1}{\beta}t^\beta$.

f) The Legendre transform of the function $f(x) = e^x$ is the function $f^*(t) = t \ln \frac{t}{e}, t > 0$, and the inequality $xt \le e^x + t \ln \frac{t}{e}$ holds for $x \in \mathbb{R}$ and $t > 0$.

Solution. a) Condition 1. I is bounded, but f is unbounded, or I and f are unbounded, but $|f|$ grows faster then px.
Condition 2. I and f are unbounded, and f grows as fast as px, for some p.
Condition 3. I is

7 Basic Inequalities

Mean Vallue Inequalitie chain

$$H_n \le G_n \le A_n \le Q_n$$

$$(\sum_{i=1}^{n} \alpha_i x_i^{-1})^{-1} \le \prod_{i=1}^{n} x_i^{\alpha_i} \le \sum_{i=1}^{n} \alpha_i x_i \le (\sum_{i=1}^{n} \alpha_i x_i^2)^2$$

Common

$$\frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}} \le \sqrt[n]{\prod_{i=1}^{n} x_i} \le \frac{\sum_{i=1}^{n} x_i}{n} \le \sqrt{\frac{\sum_{i=1}^{n} x_i^2}{n}}$$

(Developed) Bernoulli's Inequality

$$x^\alpha - \alpha(x-1) - 1 \ge 0 \quad 0 < \alpha < 1,\ x > 0$$

$$x^\alpha - \alpha(x-1) - 1 \le 0 \quad \alpha < 0,\ \alpha > 1,\ x > 0$$

$$\Downarrow \quad x = \frac{a}{b},\ \alpha = \frac{1}{p},\quad \frac{1}{q} \triangleq 1 - \frac{1}{p} = 1 - \alpha$$

Special 1°. $G_2 \le A_2$: Young's Inequality

$$\prod_{i=1}^{2} x_i^{\alpha_i} \le \sum_{i=1}^{2} \alpha_i x_i \quad (\sum_{i=1}^{2} \alpha_i = 1,\ \alpha_i > 0)$$

$\Leftrightarrow$ or

$$a^{\frac{1}{p}} b^{\frac{1}{q}} \le \frac{1}{p}a + \frac{1}{q}b \quad (\frac{1}{p} + \frac{1}{q} = 1,\ p > 1)$$

$$a^{\frac{1}{p}} b^{\frac{1}{q}} \ge \frac{1}{p}a + \frac{1}{q}b \quad (\frac{1}{p} + \frac{1}{q} = 1,\ p < 1)$$

$$\Downarrow \quad x_1 = \frac{x_i^{\frac{1}{\alpha_1}}}{\sum_{i=1}^{n} x_i^{\frac{1}{\alpha_1}}},\ x_1 = \frac{y_i^{\frac{1}{\alpha_2}}}{\sum_{i=1}^{n} y_i^{\frac{1}{\alpha_2}}} \text{ then } \Sigma$$

2°. Hölder's Inequality

$$\sum_{i=1}^{n} x_i y_i \le (\sum_{i=1}^{n} x_i^{\frac{1}{\alpha_1}})^{\alpha_1} (\sum_{i=1}^{n} y_i^{\frac{1}{\alpha_2}})^{\alpha_2}$$

$\Leftrightarrow$ or

$$\sum_{i=1}^{n} x_i y_i \le (\sum_{i=1}^{n} x_i^p)^{\frac{1}{p}} (\sum_{i=1}^{n} y_i^q)^{\frac{1}{q}} \quad (\frac{1}{p} + \frac{1}{q} = 1,\ p > 1)$$

$$\sum_{i=1}^{n} x_i y_i \ge (\sum_{i=1}^{n} x_i^p)^{\frac{1}{p}} (\sum_{i=1}^{n} y_i^q)^{\frac{1}{q}} \quad (\frac{1}{p} + \frac{1}{q} = 1,\ p < 1)$$

$$\Downarrow \quad \alpha_1 = \alpha_2 = \frac{1}{2}$$

3°. Cauchy's Inequality

$$(\sum_{i=1}^{n} x_i y_i)^2 \le (\sum_{i=1}^{n} x_i^2)(\sum_{i=1}^{n} y_i^2)$$

Apply Hölder's Inequality for indentity:

$$\sum_{i=1}^{n} (x_i + y_i)^p = \sum_{i=1}^{n} x_i (x_i + y_i)^{p-1} + \sum_{i=1}^{n} y_i (x_i + y_i)^{p-1}$$

4°. Minkowski's Inequality

$$(\sum_{i=1}^{n} (x_i + y_i)^p)^{\frac{1}{p}} \le (\sum_{i=1}^{n} (x_i)^p)^{\frac{1}{p}} + (\sum_{i=1}^{n} (y_i)^p)^{\frac{1}{p}} \quad (p > 1)$$

$$(\sum_{i=1}^{n} (x_i + y_i)^p)^{\frac{1}{p}} \ge (\sum_{i=1}^{n} (x_i)^p)^{\frac{1}{p}} + (\sum_{i=1}^{n} (y_i)^p)^{\frac{1}{p}} \quad (p < 1,\ p \ne 0)$$

Figure 2: Relations between basic inequalities

8 Multiple Integrals Short Sheet (Calculus)

8.1 Development of 1-dim Reimann Integral-Line Integral

8.1.1 Intrduction (ds)

Introduction

$I = \int_C f(x,y)\,ds,\ C: r = r(\theta),\ \theta \in [\theta_1, \theta_2]$

$C: \begin{cases} x = r(\theta)\cos\theta \\ y = r(\theta)\sin\theta \end{cases},\ f(x,y) = f(r\cos\theta, r\sin\theta)$

$ds = \sqrt{x'(\theta)^2 + y'(\theta)^2}d\theta = \sqrt{r'^2 + r^2}d\theta$

$I = \int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r'^2 + r^2}\,d\theta$

8.1.2 I-Line Integral

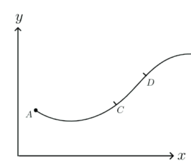
Definition

$I = \int_C f(x,y)\,ds = \left[\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}\right] = \int_{D_t} f(\varphi(t), \psi(t))\sqrt{\varphi'(t)^2 + \psi'(t)^2}\,dt$

A mean value property

$\rho(x,y) = f(x,y),\ \int_{C(A,B)} f(x,y)\,ds = M,\ \int_{C(C,D)} f(x,y)\,ds = m$

$\frac{\int_{C(C,D)} f(x,y)\,ds}{\int_{C(A,B)} f(x,y)\,ds} = \frac{\int_{C(C,D)} C\,ds}{\int_{C(A,B)} C\,ds} = \frac{S_{CD}}{S_{AB}}$



8.1.3 II-Line Integral

Definition

$I = \int_C f(x,y)\,dx = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t))\varphi'(t)\,dt$

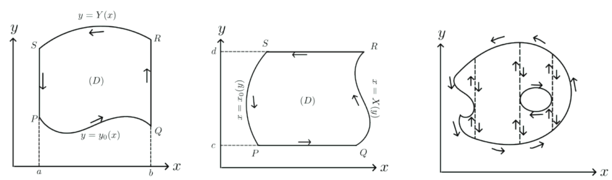
$I = \int_C f(x,y)\,dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t))\psi'(t)\,dt$

$I = \int_C P\,dx + Q\,dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} [P\varphi' + Q\psi']\,dt$

8.1.4 Calculate area using II-Line Integral

Calculate

$D = -\int_C y\,dx = \int_C x\,dy = \frac{1}{2}\int_C x\,dy - y\,dx$, where C is a simple closed cure with positive direction.



8.1.5 Conditions for curve integrals to be independent of the path

Theorem

In simply connected space D and given a path $C : A \to B$, II-line integral $\int_C P\,dx + Q\,dy$ is dependent of the integration path $C : A \to B$ i.f.f. $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Proof idea

Necessity "$\Rightarrow$":

By studying the differentiation method (partial differentiation with respect to x and y) of $F(x, y) = \int_{(x_0,y_0)}^{(x,y)} Pdx + Qdy$ and the primitive function of $Pdx+Qdy$ (regarding it as the total differential of some function) (i.e., $\exists\Phi$ such that $P = \frac{\partial \Phi}{\partial x}$, $Q = \frac{\partial \Phi}{\partial y}$ and $\int_C Pdx + Qdy = \Phi(x, y)\Big|_{(x_A,y_A)}^{(x_B,y_B)}$), we can conclude that the necessary and sufficient condition for $\int_C Pdx+Qdy$ to be path-independent is that $Pdx+Qdy$ $(= d\Phi(x))$ is an exact differential. The latter implies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ (necessity).

Sufficiency "$\Leftarrow$":

Fix $y \in [c, d]$, integrate $P = \frac{\partial \Phi}{\partial x}$ on $[x_0, x] \subset D$, then $\Phi(x, y) = \int_{x_0}^{x} Pdx + \Phi(x_0, y)$. Also, fix $x = x_0$, integrate $Q = \frac{\partial \Phi}{\partial y}$ on $[y_0, y] \subset D$, then $\Phi(x_0, y) = \int_{y_0}^{y} Q(x_0, y)dy + \Phi(x_0, y_0)$.

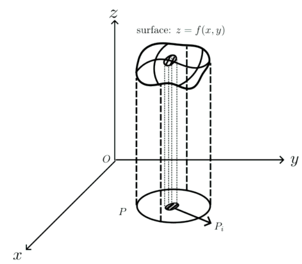
Then $\Phi(x, y) = \int_{x_0}^{x} P(x, y)dx + \int_{y_0}^{y} Q(x_0, y)dy + \Phi(x_0, y_0)$ or $\Phi(x, y) = \int_{y_0}^{y} Q(x, y)dy + \int_{x_0}^{x} P(x, y_0)dx + \Phi(x_0, y_0)$, which satisfies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

8.2 2-dim Multiple Integral

8.2.1 Definition

Definition

$\iint_P f(x,y)dP = \lim\limits_{\lambda=\max_i\{P_i\}\to 0} \sum\limits_i f(\xi_i,\eta_i)\Delta P_i = V$



8.2.2 Calculate: 2-dim Multiple Integral → Iterated Integral

2-dim Multiple Integral → Iterated Integral

Rectangle $P = [a,b]\times[c,d]$:

If $\iint_P f(x,y)dP$ exists and $I_1 = \int_a^b f(x,y)dx$ exists $\forall y \in [c,d]$, then $\iint_P f(x,y)dP = \int_c^d dy \int_a^b f(x,y)dx$.

If $\iint_P f(x,y)dP$ exists and $I_2 = \int_c^d f(x,y)dy$ exists $\forall x \in [a,b]$, then $\iint_P f(x,y)dP = \int_a^b dx \int_c^d f(x,y)dy$.

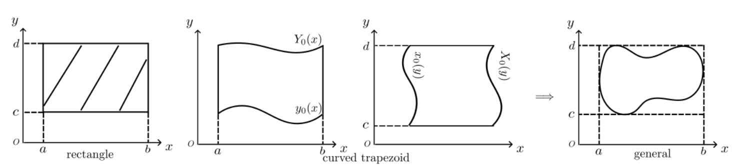
$\Longrightarrow$ If I_1 and I_2 exist, then $\iint_P f(x,y)dP = \int_a^b dx \int_c^d f(x,y)dy = \int_c^d dy \int_a^b f(x,y)dx$.

Curved trapezoid $P = [a,b]\times[y_0(x),Y(x)]$ or $P = [x_0(y),X(y)]\times[c,d]$:

If $\iint_P f(x,y)dP$ exists and $I_2 = \int_{y_0(x)}^{Y(x)} f(x,y)dy$ exists $\forall x \in [a,b]$, then $\iint_P f(x,y)dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x,y)dy$.

If $\iint_P f(x,y)dP$ exists and $I_1 = \int_{x_0(y)}^{X(y)} f(x,y)dx$ exists $\forall y \in [c,d]$, then $\iint_P f(x,y)dP = \int_c^d dy \int_{x_0(y)}^{X(y)} f(x,y)dx$.

$\Longrightarrow$ If I_1 and I_2 exist, then $\iint_P f(x,y)dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x,y)dy = \int_{y_0(x)}^{Y(x)} dy \int_a^b f(x,y)dx$.



8.2.3 Green's Formula: 2-dim Multiple Integral $\longleftrightarrow$ II-Line Integral

Green's Formula

$$\iint_D \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y}\right) dxdy = \oint_{C=\partial D} P\,dy + Q\,dx$$

or

$$\iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dxdy = \oint_{C=\partial D} P\,dy - Q\,dx$$

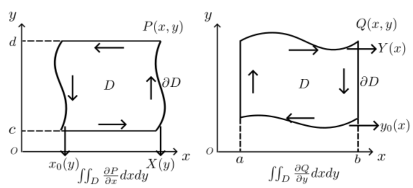
where D is a simply connected domain in $\mathbb{R}^2$ with boundary $C = \partial D$ and P, Q are continuously differentiable functions on D.

Proof idea

$P, Q, \frac{\partial P}{\partial x}, \frac{\partial Q}{\partial y}$ continuously on D, $\partial D \triangleq C$.

$\iint_D \frac{\partial P}{\partial x} dxdy = \int_c^d dy \int_{x_0(y)}^{X(y)} \frac{\partial P}{\partial x} dx = \int_c^d [P(X(y), y) - P(x_0(y), y)]dy = \oint_{\partial D} P(x, y)dy$

$\iint_D \frac{\partial Q}{\partial y} dxdy = \int_a^b dx \int_{y_0(x)}^{Y(x)} \frac{\partial Q}{\partial y} dy = \int_a^b [Q(x, Y(x)) - Q(x, y_0(x))]dx = -\oint_{\partial D} Q(x, y)dx$



8.2.4 Variable Substitution in Double Integrals

Intro

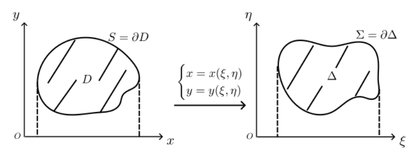
$D = \oint_{S=\partial D} xdy = \left[\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}\right] = \oint_{\Sigma=\partial\Delta} x(\xi, \eta)\frac{\partial y}{\partial \xi} d\xi + x(\xi, \eta)\frac{\partial y}{\partial \eta} d\eta = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$

Analysis

As we could see, the essential of $\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}$ actually is coordinate transform between the x, y-plane and the ξ, η-plane.

$D = \oint_S xdy = \left[\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}, \begin{cases} \xi = \xi(t) \\ \eta = \eta(t) \end{cases}\right] = \int_\alpha^\beta x(\xi(t), \eta(t))(\frac{\partial y}{\partial \xi}\xi'(t) + \frac{\partial y}{\partial \eta}\eta'(t))dt = \oint_\Sigma x(\xi, \eta)\frac{\partial y}{\partial \xi} d\xi + x(\xi, \eta)\frac{\partial y}{\partial \eta} d\eta$, let $x(\xi, \eta)\frac{\partial y}{\partial \xi} \triangleq P(x, y)$, $x(\xi, \eta)\frac{\partial y}{\partial \eta} \triangleq Q(x, y)$, then $D = \oint_\Sigma P d\xi + Q d\eta = (\frac{\partial P}{\partial \eta} = \frac{\partial Q}{\partial \xi}$, using Greeni's formula$) = \iint_\Delta (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi d\eta = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$.

Figures for aforementioned analysis



Idea of Analysis:

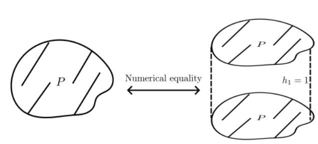
Using $\begin{cases} x = x(\xi,\eta) = x(\xi(t),\eta(t)) \\ y = y(\xi,\eta) = y(\xi(t),\eta(t)) \end{cases}$ (coordinate transform with parameterization), transform the II-line integration $\oint_S xdy$ (or $-\oint_S ydx$, or $\frac{1}{2}\oint_S xdy - ydx$) on x, y-plane to the II-line integral $\oint_\Sigma P(\xi,\eta)d\xi + Q(\xi,\eta)d\eta$, then using Green's formula to transform it to double integral $\iint_\Delta (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi d\eta$ on ξ, η-plane, following is how we calculate the double integral (often transform it to iterated integral $\int_{D\xi} d\xi \int_{\eta_0(\xi)}^{\eta(\xi)} (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi$, or others).

/Shortly, calculate the area of D on x, y plane: coordinate transform with parameterization $\rightarrow$ Green's formula $\rightarrow$ Double integration calculate/

Comparison:

Recaping the previous $D = \frac{1}{2}\oint_S xdy - ydx$, the original calculation method for this II-line integral is that find thier parametric equation $\begin{cases} x = x(t) \\ y = y(t) \end{cases}$, $t \in [a, b]$ for different (closed) curve S, then substitute into the calculation.From here, we could see that the essense of II-line integral is definite integral and has chirality.

However, "coordinate transform + Green's formula" method: $D = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$ as long as we master the coordinate transformation of a type of shape equation, we can easily calculate the results. Although there is a parameterization in the derivation, it is not reflected in the result (it can be seen from the process that if the parameterization is continued, it will be the result of the II-line integral in ξ, η-plane). In essence, the area is calculated as the volume with the same value, similiar with $\frac{1}{2}\oint_S xdy - ydx = \iint_D 1dxdy$ or $\oint_S xdy = \iint_D 1dxdy$ (If you unfold the left and right sides under the curved trapezoid, you can find the consistency.), it's just that the physical meanings of the two equations are different.



/Note: The choice of which method should be based on which is more convenient, parametric equations or coordinate transformations./

Variable Substitution in Double Integral

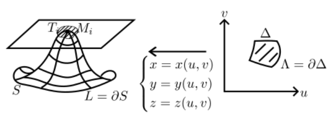
$$\iint_D f(x,y)dxdy = \begin{cases} x = x(\xi,\eta) \\ y = y(\xi,\eta) \end{cases} = \iint_\Delta f(x(\xi,\eta), y(\xi,\eta))|\frac{D(x,y)}{D(\xi,\eta)}|d\xi d\eta$$

8.3 Surface Integral

8.3.1 Definition and basics

Definition

$S = \lim\limits_{\lambda\to 0} \sum\limits_i T_i$, where T_i is the projection area of S_i on the tangent plane at point $M_i \in S_i$.



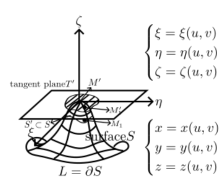
Calculte Surface Integral(Omit matrix-vector operations)

$$S = \lim_{\lambda\to 0} \sum_i T_i = \iint_\Delta \sqrt{A^2+B^2+C^2}dudv = \iint_\Delta \sqrt{EG-F^2}dudv$$

where $A = \begin{vmatrix} y'_u & z'_u \\ y'_v & z'_v \end{vmatrix}$, $B = \begin{vmatrix} z'_u & x'_u \\ z'_v & x'_v \end{vmatrix}$, $C = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix}$, $E = \sum\limits_{x,y,z} x'^2_u$, $F = \sum\limits_{x,y,z} x'_u x'_v$, $G = \sum\limits_{x,y,z} x'^2_v$.

Procedure:

$\forall M_1 \in S'$, the coordinates of its projection M'_1 on T' are $\begin{cases} \xi = \xi(u,v) \\ \eta = \eta(u,v) \\ \zeta = \zeta(u,v) \end{cases}$, then $S'_T = \iint_\Delta |\frac{D(\xi,\eta)}{D(u,v)}|dudv$, then the results could appear after a series of matrix-vector operations.



8.3.2 Surface Integral

I-Surface Integral

$$\begin{aligned}
I &= \iint_S f(x,y,z)\,ds \\
&= [x = x(u,v), y = y(u,v), z = z(u,v), /parameterization/ \\
&\quad dS = \sqrt{A^2+B^2+C^2}\,du\,dv = \sqrt{EG-F^2}\,du\,dv] \\
&= \iint_\Delta f(x(u,v),y(u,v),z(u,v))\sqrt{A^2+B^2+C^2}\,du\,dv \\
&= \iint_\Delta f(x(u,v),y(u,v),z(u,v))\sqrt{EG-F^2}\,du\,dv
\end{aligned}$$

Special case:
If $u = x,\ v = y,\ z = z(x,y)$,

$$\begin{aligned}
I &= \iint_\Delta f(x,y,z(x,y))\sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy \\
&= \iint_\Delta f(x,y,z(x,y))\frac{dx\,dy}{|\cos(v)|}
\end{aligned}$$

II-Surface Integral

$$I = \iint_S f(x,y,z)\,dx\,dy = \text{or} \iint_S f(x,y,z)\,dy\,dz = \text{or} \iint_S f(x,y,z)\,dz\,dx$$
$$\text{or} \iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy$$

Connection between I and II-Surface Integrals

$$\underset{(II)}{\iint_S f(x,y,z)\,dx\,dy} = \underset{(I)}{\iint_S f(x,y,z)\cos(v)\,ds}$$

$$\begin{aligned}
&\text{(Calculate it: parameterize to get double integral)} \\
&= [x = x(u,v), y = y(u,v), z = z(u,v), /parameterize/ \\
\cos(v) = \pm\frac{C}{\sqrt{A^2+B^2+C^2}},&\ dS = \sqrt{A^2+B^2+C^2}\,du\,dv] \\
&= \pm\iint_\Delta f(x(u,v),y(u,v),z(u,v))\cdot C\,du\,dv
\end{aligned}$$

or

$$\underbrace{\iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy}_{(II)} = \underbrace{\iint_S P\cos(\lambda) + Q\cos(\mu) + R\cos(\nu)\,dS}_{(I)}$$

(Calculate it: parameterize to get double integral)

$$= [\begin{cases} x = x(u,v) \\ y = y(u,v) \quad /parameterize/ \\ z = z(u,v) \end{cases}$$

$$\cos(\lambda) = \pm\frac{A}{\sqrt{A^2+B^2+C^2}}, \cdots,$$

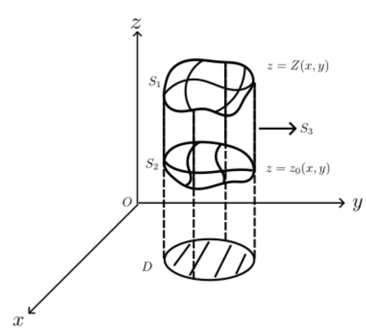
$$dS = \sqrt{A^2+B^2+C^2}\,du\,dv]$$

$$= \pm\iint_\Delta [P\cdot A + Q\cdot B + R\cdot C]\,du\,dv$$

8.3.3 Calculate Volume using Surface Integral

Calculate Volume using Surface Integral

$$\begin{aligned} V &= \iint_D Z(x,y)\,dx\,dy - \iint_D z_0(x,y)\,dx\,dy \\ &= \iint_{S_2} z\,dx\,dy + \iint_{S_1} z\,dx\,dy \\ &= \iint_S z\,dx\,dy \ (S = S_1 + S_2 + S_3, \text{ outside}) \\ or &= \iint_S x\,dy\,dz \\ or &= \iint_S y\,dz\,dx \\ or &= \frac{1}{3}\iint_S (x\,dy\,dz + y\,dz\,dx + z\,dx\,dy) \end{aligned}$$



8.3.4 Stoke's Formula: II-Surface Integral $\longleftrightarrow$ II-Line Integral

Stoke's Formula $\Leftarrow$ Development of Green's Formula
$\int_{L=\partial S} P\,dx + Q\,dy + R\,dz = \iint_S (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\,dx\,dy + (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z})\,dy\,dz + (\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x})\,dz\,dx$

Proof idea
Using Green's formula on LHS as parameterizing $[x = x(u,v), y = y(u,v), z = z(u,v)]$, it's easy to prove.

8.4 3-dim Multiple Integral

8.4.1 Definition and basics

Definition
$\iiint_V f(x,y,z)\,dV = \lim_{\lambda\to 0} \sum_i f(x_i, y_i, z_i)\Delta V_i$.

Calculate

$$\begin{aligned}
\iiint_V f(x,y,z)\,dV &= \int_a^b dx \iint_{P_y z} f(x,y,z)\,dy\,dz \\
&(\iint_{P_y z} f(x,y,z)\,dy\,dz\ \exists) \\
&= \int_a^b dx \int_c^d dy \int_e^f f(x,y,z)\,dz \\
&(\int_e^f f(x,y,z)\,dz\ \exists) \\
\text{also } &= \iint_{P_x y} dx\,dy \int_e^f f(x,y,z)\,dz
\end{aligned}$$

$$\begin{aligned}
\iiint_V f(x,y,z)\,dV &= \int_{x_0}^X dx \iint_{P_y z} f(x,y,z)\,dy\,dz \\
&(\iint_{P_y z} f(x,y,z)\,dy\,dz\ \exists) \\
&= \int_{x_0}^X dx \int_{y_0}^{Y(x)} dy \int_{z_0(x,y)}^{Z(x,y)} f(x,y,z)\,dz \\
&(\int_{z_0(x,y)}^{Z_(x,y)} f(x,y,z)\,dz\ \exists) \\
\text{also } &= \iint_{P_x y} dx\,dy \int_{z_0(x,y)}^{Z_(x,y)} f(x,y,z)\,dz
\end{aligned}$$

8.4.2 Gauss's Formual 3-dim Multiple Integral $\longleftrightarrow$ II-Surface Integral

Gauss's Formula: $\Leftarrow$ Dimension-increasing version of Stokes' formula

$$\begin{cases} \iiint_V \frac{\partial R}{\partial z}\,dx\,dy\,dz = \iint_{S=\partial V} R\,dx\,dy \\ \iiint_V \frac{\partial P}{\partial x}\,dy\,dz\,dx = \iint_{S=\partial V} P\,dy\,dz \\ \iiint_V \frac{\partial Q}{\partial y}\,dz\,dx\,dy = \iint_{S=\partial V} Q\,dz\,dx \end{cases}$$

$\Sigma\ :\ \Longrightarrow\quad \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz \ = \ \iint_{S=\partial V} P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy$ (S is alonge the outside)

8.4.3 Variable Substitution in 3-dim Multiple Integral

Intro

$$V = \iint_S z\,dx\,dy$$

(coordinate transform)

$$= \iint_E z \cdot \frac{D(x,y)}{D(u,v)}\,du\,dv$$

$$= \iint_E z \cdot \left[\frac{D(x,y)}{D(\xi,\eta)}\frac{D(\xi,\eta)}{D(u,v)} + \frac{D(x,y)}{D(\eta,\zeta)}\frac{D(\eta,\zeta)}{D(u,v)} + \frac{D(x,y)}{D(\zeta,\xi)}\frac{D(\zeta,\xi)}{D(u,v)}\right] du\,dv$$

(II-Surface Integral) (using Green's formula)

$$= \iint_\Sigma z\frac{D(x,y)}{D(\xi,\eta)}\,d\xi\,d\eta + z\frac{D(x,y)}{D(\eta,\zeta)}\,d\eta\,d\zeta + z\frac{D(x,y)}{D(\zeta,\xi)}\,d\zeta\,d\xi$$

$$= \iiint_\Delta \left(\frac{\partial}{\partial \zeta}\left[z\frac{D(x,y)}{D(\xi,\eta)}\right] + \frac{\partial}{\partial \xi}\left[z\frac{D(x,y)}{D(\eta,\zeta)}\right] + \frac{\partial}{\partial \eta}\left[z\frac{D(x,y)}{D(\zeta,\xi)}\right]\right) d\xi\,d\eta\,d\zeta$$

(rearrange)

$$= \iint_\Delta |\frac{D(x,y,z)}{D(\xi,\eta,\zeta)}|\,d\xi\,d\eta\,d\zeta$$

Variable Substitution in 3-dim Multiple Integral

$$\iiint_D f(x,y,z)\,dx\,dy\,dz$$

$$= \left[\begin{cases} x = x(\xi,\eta,\zeta) \\ y = y(\xi,\eta,\zeta) \\ z = z(\xi,\eta,\zeta) \end{cases} \quad /parameterize/ \right]$$

$$= \iiint_\Delta f(x(\xi,\eta,\zeta), y(\xi,\eta,\zeta), z(\xi,\eta,\zeta)) \cdot |\frac{D(x,y,z)}{D(\xi,\eta,\zeta)}|\,d\xi\,d\eta\,d\zeta$$

8.5 n-dim Multiple Integral

8.5.1 Definition and basics

Definition

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n = \lim_{\lambda(P)\to 0} \sigma(f, P, \xi)$$

where I is an n-dimensional interval, P is a partition of I, $\lambda(P)$ is the measure of P, and $\sigma(f, P, \xi)$ is the Riemann sum.

Calculate

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n$$

(Special)

$$= \int_{a_1}^{b_1} dx^1 \int_{a_2}^{b_2} dx^2 \cdots \int_{a_n}^{b_n} f(x^1, \cdots, x^n)\, dx^n$$

(General)

$$= \int_{x_1}^{X_1} dx^1 \int_{x_2(x_1)}^{X_2(x_1)} dx^2 \cdots \int_{x_n(x_1,\cdots,x_{n-1})}^{X_n(x_1,\cdots,x_{n-1}} f(x^1, \cdots, x^n)\, dx^n$$

8.5.2 Variable Substitution in n-dim Multiple Integral

Variable Substitution in n-dim Multiple Integral

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n$$

$$= \left[\begin{cases} x^1 = x^1(\xi^1, \cdots, \xi^n) \\ x^2 = x^2(\xi^1, \cdots, \xi^n) \\ \cdots \\ x^n = x^n(\xi^1, \cdots, \xi^n) \end{cases} \quad /parameterize/ \right]$$

$$= \int_\Delta f(x^1(\xi^1, \cdots, \xi^n), \cdots, x^n(\xi^1, \cdots, \xi^n)) |\frac{D(x)}{D(\xi)}|\, d\xi^1 \cdots d\xi^n$$

where $\frac{D(x)}{D(\xi)} = \frac{D(x^1,\cdots,x^n)}{D(\xi^1,\cdots,\xi^n)}$ is the Jacobian determinant of the transformation.

9 Multiple Integrals

9.1 The Riemann Integral over an n-Dimensional Interval

9.1.1 Comprehensions

First, the definition of Riemann integral over n-dimention interval is simimiar (almost same) with 1-dimension condition ($\int_I f(x)dx := \lim\limits_{\lambda(P)\to 0} \sigma(f, P, \xi)$),

$$\int_I f(x^1, \cdots, x^n)dx^1 \cdots dx^n \text{ or } \int \cdots \int_I f(x^1, \cdots, x^2)dx^1 \cdots dx^n$$

Then, introduce the necessity condition of it ($f \in \mathcal{R}(I) \implies f$ is bounded on I). After that, book gives some properties (Lemmas) about the set of measure zero (same with 1-dimension condition), and developes the Cantor Theorem (about uniform continuity of continuous functions on compact set). At last, we're given 2 criterions (necessity and sufficiency) of the Riemann integral (Darboux Criterion, Lebesgue Criterion).

Example. Let $f : I \to \mathbb{R}$ is a continuous real-value function on $I \subset \mathbb{R}^{n-1}$, then it's graph is a set measure zero in n-dimension.

Lemma 9.1. */Development of Cantor Theorem/ If the relation $\omega(f; x) < w_0$ holds at each point of a compact set K for the function $f : K \to \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\omega(f; U_\delta(x)) < w_0 + \varepsilon$ for each point $x \in K$.*

Theorem 9.1. *(Lebesgue's criterion). $f \in \mathcal{R}(I) \Leftrightarrow (f$ is bounded on $I) \wedge (f$ is continuous almost everywhere on $I)$.*

Theorem 9.2. *(Darboux Criterion) $f \in (I) \Leftrightarrow (f$ is bounded on $I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup\limits_P s(f, P)$, $\overline{\mathcal{J}} = \inf\limits_P S(f, P)$.*

9.1.2 Problems Sets

Problem 53. a) Show that a set of measure zero has no interior points.

b) Show that not having interior points by no means guarantees that a set is of measure zero.

c) Construct a set having measure zero whose closure is the entire space $\mathbb{R}^n$.

d) A set $E \subset I$ is said to have content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \ldots, I_k$ such that $\sum_{i=1}^{k} |I_i| < \varepsilon$. Is every bounded set of measure zero a set of content zero?

e) Show that if a set $E \subset \mathbb{R}^n$ is the direct product $\mathbb{R} \times e$ of the line $\mathbb{R}$ and a set $e \subset \mathbb{R}^{n-1}$ of $(n-1)$-dimensional measure zero, then E is a set of n-dimensional measure zero.

Proof. a) [Proof by contradiction] If E has a point x_0 as an interior point, then there exists a ball $B(x_0, r)$ such that $B(x_0, r) \subset E$. Since the ball has positive measure, it follows that E cannot have measure zero.

b) [Counterexample] Consider the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$, which have no interior points in $\mathbb{R}$ but are not of measure zero.

c) /inspired by b)/ Consider the set $E = \mathbb{Q}^n$, the set of all points in $\mathbb{R}^n$ with rational coordinates. The closure of E is $\mathbb{R}^n$, and E has measure zero since it can be covered by countably many intervals whose total length can be made arbitrarily small.

d) [Principle] No, since the Lebesgue measure zero set can be covered by at most countable intervals (which means measure $\mu = o(1) \cdot \mathcal{O}(1)$), while a set of content zero can be covered by a finite number of intervals (which means the content (also a kind of measure) sometimes is $\mathcal{O}(1) \cdot \mathcal{O}(1)$). [Example] The $\mathbb{Q} \cap [0, 1]$ and Cantor set are sets of measure zero but none of content zero.

e) From the construction of the product measure λ_n (Lebesgue measure of $\mathbb{R}^n$) /Lemma1/ if $A \in B(\mathbb{R}^k)$ and $B \in B(\mathbb{R}^{n-k})$, $(0 < k < n)$, then $\lambda_n(A \times B) = \lambda_k(A)\lambda_{n-k}(B)$. So here (using also that /Lemma2/ if $E_m \uparrow E$, $\lim_{m\to+\infty} \lambda_n(E_m) = \lambda_n(E)$) $\lambda_n(E) = \lambda_n(\mathbb{R}\times e)$ $= \lim_{m\to+\infty} \lambda_n((-m, m) \times e) = \lim_{m\to+\infty} \lambda_1((-m, m)) \times \lambda_{n-1}(e)$ $= \lim_{m\to+\infty} \lambda_1((-m, m)) \times 0 = 0$.

Notes. 1. The proof of e) /Lemma2/ could found in *Real and Complex Analysis* by Walter Rudin, Theorem 1.19(Page 16), based on the (positive) measure definition (countable additive of disjoint countable measurable sets).

Problem 54. a) Construct the analogue of the Dirichlet function in $\mathbb{R}^n$ and show that a bounded function $f : I \to \mathbb{R}$ equal to zero at almost every point of the interval I may still fail to belong to $\mathcal{R}(I)$.

b) Show that if $f \in \mathcal{R}(I)$ and $f(x) = 0$, at almost all points of the interval I, then $\int\limits_I f(x)\,\mathrm{d}x = 0$.

Proof. a) [Counterexample] Consider $\mathcal{D}^n = \begin{cases} 1 & \text{if } x \in \mathbb{Q}^n \\ 0 & \text{if } x \in \mathbb{R}^n \backslash \mathbb{Q}^n \end{cases}$, the discontinuous set is $\mathcal{R}^n$.

b) /by definition/ $\int_I f(x)dx = \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_i \in P} f(\xi_i)\Delta(x_i) = \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_i \in A\cap P} f(\xi_i)\Delta(x_i) + \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_j \in B\cap P} f(\xi_j)\Delta(x_i) = \mathcal{O}(1) \cdot o(1) + o(1) \cdot \mathcal{O}(1)$, where $A = \{\cup[x_k, x_{k+1}] \subset P : \forall k,\ \exists \xi \in [x_k, x_{k+1}],\ s.t.\ f(\xi) \neq 0\}$ and $B = I \backslash A$.

Problem 55. There is a small difference between our earlier definition of the Riemann integral on a closed interval $I \subset \mathbb{R}$ and Definition 7 for the integral over an interval of arbitrary dimension. This difference involves the definition of a partition and the measure of an interval of the partition. Clarify this nuance for yourself and verify that

$$\int_a^b f(x)dx = \begin{cases} \int_I f(x)dx & \text{if } a < b \\ -\int_I f(x)dx & \text{if } a > b \end{cases}$$

where I is the interval on the real line $\mathbb{R}$ with endpoints a and b.

Proof. Just understand it, it's not hard.

Problem 56. a) Prove that a real-valued function $f : I \to \mathbb{R}$ defined on an interval $I \subset \mathbb{R}^n$ is integrable over that interval if and only if for every $\varepsilon > 0$ there exists a partition P of I such that $S(f; P) - s(f; P) < \varepsilon$.

b) Using the result of a) and assuming that we are dealing with a real-valued function $f : I \to \mathbb{R}$, one can simplify slightly the proof of the sufficiency of the Lebesgue criterion. Try to carry out this simplification by yourself.

Proof. a) /by The Darboux Criterion/ $f \in (I) \Leftrightarrow$ (f *is bounded on* I) $\wedge$ ($\underline{\mathcal{J}} = \overline{\mathcal{J}}$), where $\underline{\mathcal{J}} = \sup\limits_P s(f, P)$, $\overline{\mathcal{J}} = \inf\limits_P S(f, P)$. Notice that $\forall\varepsilon > 0,\ \exists P,\ s.t.\ S(f, P) - s(f, P) < \varepsilon \Leftrightarrow \underline{\mathcal{J}} = \overline{\mathcal{J}}$ (proof is similiar with 1-dimention condition), which is what we need to prove.

b) [Segmented estimation] Sufficiency: procedure is same with book, a) just simplify the last step.

9.2 The Integral over a Set

9.2.1 Comprehensions

This part is not very long and mainly revolves around the concept of "admissible set".

First, the book introduces the concept (i.e. definition) of "admissible set" and some (operational) properties of it. Then, using a characteristic function χ_E of set E, we're given the definition of the integral over a set E and the corresponding Lebesgue's Criterion (over "admissible set"). At last, the book also gives the Jordan measure (or volume) of a bounded set E (actually it's only well defined for "admissible set").

Definition 1. *An admissible set is a bounded set $E \subset \mathbb{R}^n$ s.t. ∂E is measure zero in sense of Lebesgue.*

Lemma 9.2. *$\forall$ sets $E,\ E_1,\ E_2 \subset \mathbb{R}^n$,*

a) ∂E is the closed set on $\mathbb{R}^n$.

b) $\partial(E_1 \cup E_2),\ \partial(E_1 \cap E_2),\ \partial(E_1 \backslash E_2) \subset \partial E_1 \cup \partial E_2$.

c) Finite union or intersection of admissible sets is still admissible set, the difference of admissible sets is still admissible set.

Notes. 1. Generally, for infinite admissible sets, Lemma 2.2 c) is not true.
2. The boundary of admissible set is compact on $\mathbb{R}^n$ (closed and bounded).

Definition 2. *The integral of f over E is given by $\int_E f(x)dx := \int_{\Gamma \supset E} f\chi_E(x)dx$, where I is any interval containing E. If the integral exists, we say $f \in \mathcal{R}(E)$.*

$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$ *is the characteristic function of set E and* $f_{\chi_E}(x) = \begin{cases} f(x) & x \in E \\ 0 & x \notin E \end{cases}$.

Notes. 1. $\chi_E(x)$ is interrupted only on the boundary ∂E. The points of discontinuity of $f_{\chi_E}(x)$ are either points of discontinuity of f on E, or the result of discontinuities of χ_E, in which case they lie on ∂E.
2. $\forall I_1,\ I_2 \supset E$, integrals $\int_{I_1} f_{\chi_E}(x)dx$ and $\int_{I_2} f_{\chi_E}(x)dx$ either both exist (then values are equal) or both fail to exist.

Theorem 9.3. *A function $f : E \to \mathbb{R}$ is integrable over an admissible set if and only if it is bounded and continuous at almost all points of E.*

Notes. 1. Compared with f, f_{χ_E} may have additional points of discontinuity only on the ∂E.

Definition 3. *The (Jordan) measure or content of a bounded set $E \subset \mathbb{R}^n$ is $\mu(E) := \int_E 1 \cdot dx = \int_{I \supset E} \chi_E(x)dx$, provided this Riemann integral exists.*

Notes. 1. Only admissible sets, are measurable in the sense of aformentioned definition.

Comprehensions for Definition 3(From Book.)

The geometric meaning of $\mu(E)$ (if E is admissible) could be seen in following way:

$$\mu(E) = \int_{I \supset E} \chi_E(x)dx = \underline{\int}_{I \supset E} \chi_E(x)dx = \overline{\int}_{I \supset E} \chi_E(x)dx$$

Due to Darboux Criterion and Theorem, $\mu(E)$ is the common limit as $\lambda(P) \to 0$ of the volumes of polyhedra inscribed in and circumscribed about E, in agreement with the accepted idea of the volume of simple solids $E \subset \mathbb{R}^n$.

Why the $\mu(E)$ is called Jordan measure?

Definition 4. *A set $E \subset \mathbb{R}^n$ is a set of measure zero in the sense of Jordan or a set of content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \ldots, I_k$ s.t. $\sum_{i=1}^k |I_i| < \varepsilon$.*

Compared with measure zero in the sense of Lebesgue, a requirement that the covering be finite appears here, shrinking the class of sets of Lebesgue measure zero. For example, the set of rational points is a set of measure zero in the sense of Lebesgue, but not in the sense of Jordan.

In order for the least upper bound of the contents of polyhedra inscribed in a bounded set E to be the same as the greatest lower bound of the contents of polyhedra circumscribed about E (and to serve as the measure $\mu(E)$ or content of E), it is obviously necessary and sufficient that the boundary ∂E of E have measure 0 in the sense of Jordan. That is the motivation for the following definition.

Definition 5. *A set E is Jordan-measurable if it is bounded and its boundary ∂E has Jordan measure zero.*

Hence, the class of Jordan-measurable subsets is precisely the class of admissible sets. That's the reason the measure $\mu(E)$ defined earlier can be called (and is called) the Jordan measure of the (Jordan-measurable) set E.

10 Appendix

10.1 /Series/ A simple test for nth term of a series to approach 0

/Dev/ /Tec Tool/

Using Stirling's formula, one may see at once that if $a_n = \frac{(2n)!}{4^n(n!)^2}$, then a_n is of the order of $\frac{1}{\sqrt{n}}$, and one may conclude from the alternating series test that the series $\sum(-1)^n a_n$ is conditionally convergent.
At an elementary level, however, the convergence of the latter series may be a little more difficult to obtain. Since $\frac{a_{n+1}}{a_n} = \frac{(2n+1)}{(2n+2)} < 1$ for each n, it is clear that the sequence $\{a_n\}$ is decreasing, but it is not immediately obvious within the environment of a typical calculus course that $a_n \to 0$ as $n \to \infty$. For this purpose, one might use the following simple result which takes a leaf out of the theory of infinite products:

Lemma. Suppose (a_n) is a decreasing sequence of positive numbers and for each natural number n, define $b_n = 1 - \frac{a_{n+1}}{a_n}$. Then the sequence $\{a_n\}$ converges to zero if and only if the series $\sum b_n$ diverges.

Proof. We note first that unless $b_n \to 0$ as $n \to \infty$, both of the series $\sum b_n$ and $\sum \log(1 - b_n)$ diverge. On the other hand, if $b_n \to 0$, then $b_n/(-\log(1-b_n)) \to 1$ as $n \to \infty$, and it follows from the limit comparison test that $\sum b_n$ diverges if and only if $\sum \log(1 - b_n)$ diverges. We note also that since $0 < b_n < 1$, we have $\log(1 - b_n) < 0$ for each n.

Now since $1 - b_n = a_{n+1}/a_n$ for every n, it is clear that $a_n = a_1(1 - b_1)(1 - b_2)(1 - b_3) \cdots (1 - b_{n-1})$ for each $n \geq 2$, and we therefore conclude that $a_n \to 0$ iff $\log a_n \to -\infty$ iff $\log a_1 + \sum_{i=1}^{n} \log(1 - b_i) \to -\infty$ iff $\sum \log(1 - b_n)$ diverges iff $\sum b_n$ diverges.
Returning now to the above example, we see that $b_n = 1/(2n + 2)$ for each n, and the obvious divergence of $\sum b_n$ implies that $a_n \to 0$. The same technique gives an easy proof of the convergence of such series as $\sum((-1)^n n^n/e^n n!)$, and the series $\sum\binom{\alpha}{n}$ of binomial coefficients with $\alpha > -1$.

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